

Module 2

AM Concepts

In AM, the information signal varies the amplitude of the carrier sine wave. The instantaneous value of the carrier amplitude changes in accordance with the amplitude and frequency variations of the modulating signal.

Fig. 1.1 shows a single-frequency sine wave intelligence signal modulating a higher-frequency carrier. The carrier frequency remains constant during the modulation process, but its amplitude varies in accordance with the modulating signal. An increase in the amplitude of the modulating signal causes the amplitude of the carrier to increase. Both the positive and the negative peaks of the carrier wave vary with the modulating signal. An increase or a decrease in the amplitude of the modulating signal causes a corresponding increase or decrease in both the positive and the negative peaks of the carrier amplitude.

An imaginary line connecting the positive peaks and negative peaks of the carrier waveform (the dashed line in Fig. 1-1) gives the exact shape of the modulating information signal. This imaginary line on the carrier waveform is known as the *envelope*.

Because complex waveforms such as that shown in Fig. 1-1 are difficult to draw, they are often simplified by representing the high-frequency carrier wave as many equally spaced vertical lines whose amplitudes vary in accordance with a modulating signal, as in Fig. 1-2.

The signals illustrated in Figs. 1-1 and 1-2 show the variation of the carrier amplitude with respect to time and are said to be in the time domain. Time-domain signals—voltage or current variations that occur over time

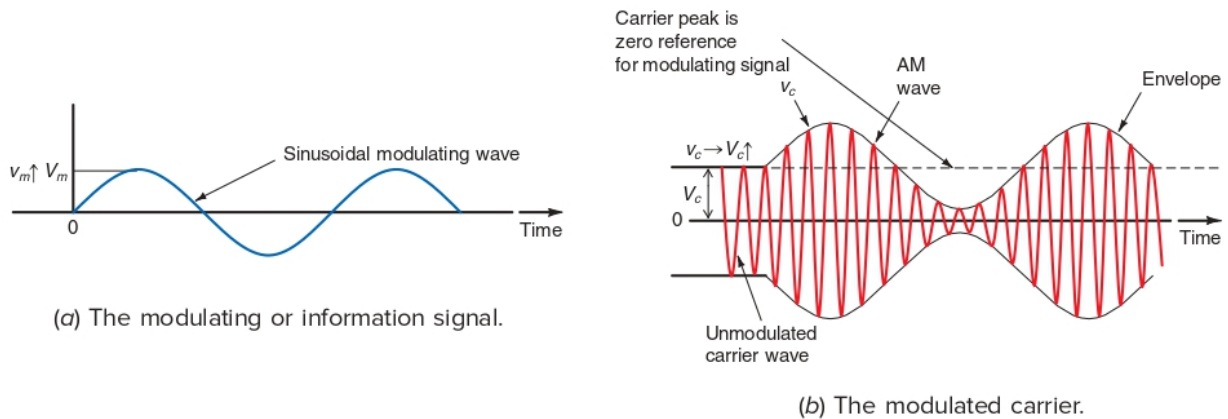


Figure 1.1 Amplitude Modulation

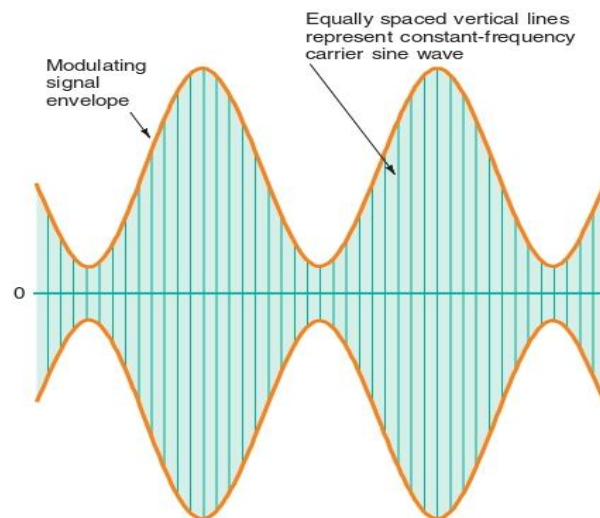


Figure 1.2 A simplified method of representing an AM high-frequency sine wave.

Using trigonometric functions, we can express the sine wave carrier with the simple expression

$$v_c = V_c \sin 2\pi f_c t$$

In this expression, v_c represents the instantaneous value of the carrier sine wave voltage at some specific time in the cycle; V_c represents the peak value of the constant unmodulated carrier sine wave as measured between zero and the maximum amplitude of either the positive-going or the negative-going alternations (Fig. 1-1); f_c is the frequency of the carrier sine wave; and t is a particular point in time during the carrier cycle.

A sine wave modulating signal can be expressed with a similar formula

$$v_m = V_m \sin 2\pi f_m t$$

where v_m = instantaneous value of information signal

V_m = peak amplitude of information signal

f_m = frequency of modulating signal

In Fig. 1-1, the modulating signal uses the peak value of the carrier rather than zero as its reference point. The envelope of the modulating signal varies above and below the peak carrier amplitude. That is, the zero reference line of the modulating signal coincides with the peak value of the unmodulated carrier.

In general, the amplitude of the modulating signal should be less than the amplitude of the carrier. When the amplitude of the modulating signal is greater than the amplitude of the carrier, distortion will occur, causing incorrect information to be transmitted. In amplitude modulation, it is particularly important that the peak value of the modulating signal be less than the peak value of the carrier. Mathematically

$$V_m < V_c$$

Values for the carrier signal and the modulating signal can be used in a formula to express the complete modulated wave. The peak value of the carrier is the reference point for the modulating signal; the value of the modulating signal is added to or subtracted from the peak value of the carrier.

The instantaneous value of either the top or the bottom voltage envelope v_1 can be computed by using the equation

$$v_1 = V_c + v_m = V_c + V_m \sin 2\pi f_m t$$

which expresses the fact that the instantaneous value of the modulating signal algebraically adds to the peak value of the carrier. Thus, we can write the instantaneous value of the complete modulated wave v_2 by substituting v_1 for the peak value of carrier voltage V_c as follows

$$v_2 = v_1 \sin 2\pi f_c t$$

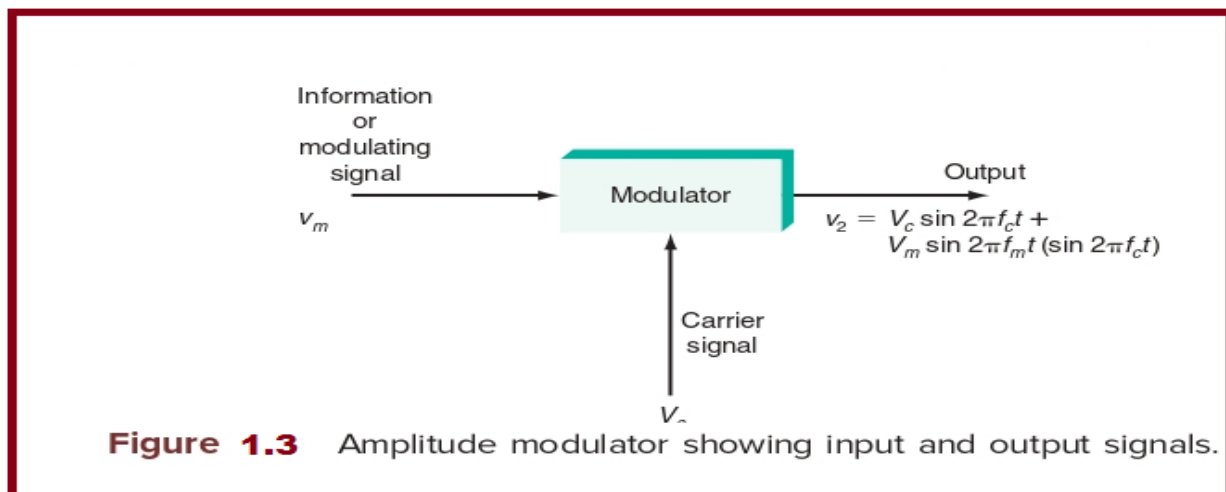
Now substituting the previously derived expression for v_1 and expanding, we get the following

$$v_2 = (V_c + V_m \sin 2\pi f_m t) \sin 2\pi f_c t = V_c \sin 2\pi f_c t + (V_m \sin 2\pi f_m t) (\sin 2\pi f_c t)$$

where v_2 is the instantaneous value of the AM wave (or v_{AM}), $V_c \sin 2\pi f_c t$ is the carrier waveform, and $(V_m \sin 2\pi f_m t) (\sin 2\pi f_c t)$ is the carrier waveform multiplied by the modulating signal waveform. It is the second part of the expression that is characteristic of AM. A circuit must be able to produce mathematical multiplication of the carrier and modulating signals in order for AM to occur. The AM wave is the product of the carrier and modulating signals.

Amplitude modulator

The circuit used for producing AM is called a *modulator*. Its two inputs, the carrier and the modulating signal, and the resulting outputs are shown in Fig. 1.3. Amplitude modulators compute the product of the carrier and modulating signals. Circuits that compute the product of two analog signals are also known as analog multipliers, mixers, converters, product detectors, and phase detectors. A circuit that changes a lower-frequency baseband or intelligence signal to a higher-frequency signal is usually called a modulator. A circuit used to recover the original intelligence signal from an AM wave is known as a detector or demodulator



Modulation Index and Percentage of Modulation

Modulation Index:

The modulating signal voltage V_m must be less than the carrier voltage V_c for undistorted AM to occur. The relationship between the amplitude of the modulating signal and the amplitude of the carrier signal is known as the *modulation index* m (also called the modulating factor or coefficient, or the degree of modulation)

Modulation index is the ratio between peak amplitude of the modulating signal to peak amplitude of carrier signal

$$m = \frac{V_m}{V_c}$$

Percentage Modulation: Multiplying the modulation index by 100 gives the *percentage of modulation*

For example, if the carrier voltage is 9 V and the modulating signal voltage is 7.5 V, the modulation factor is 0.8333 and the percentage of modulation is $0.8333 \times 100 = 83.33$.

Over modulation and Distortion

The modulation index should be a number between 0 and 1. If the amplitude of the modulating voltage is higher than the carrier voltage, m will be greater than 1, then it is called as over modulation. It causes distortion in modulated wave. If the distortion is of large amount then transmitted information signal will get distorted.

Distortion of voice transmissions produces garbled, harsh, or unnatural sounds in the speaker. Distortion of video signals produces a scrambled and inaccurate picture on a TV screen.

Simple distortion is illustrated in Fig. 1-4. Here a sine wave information signal is modulating a sine wave carrier, but the modulating voltage is much greater than the carrier voltage, resulting in a condition called *over modulation*

As shown in figure 1.4, the waveform is flattened at the zero line since A_m is larger than A_c . The received signal will produce an output waveform in the shape of the envelope, which in this case is a sine wave whose negative peaks have been clipped off. If the amplitude of the modulating signal is less than the carrier amplitude, no distortion will occur. The ideal condition for AM is when $V_m = V_c$, or $m=1$ which gives 100 percent modulation. This results in the greatest output power at the transmitter and the greatest output voltage at the receiver, with no distortion.

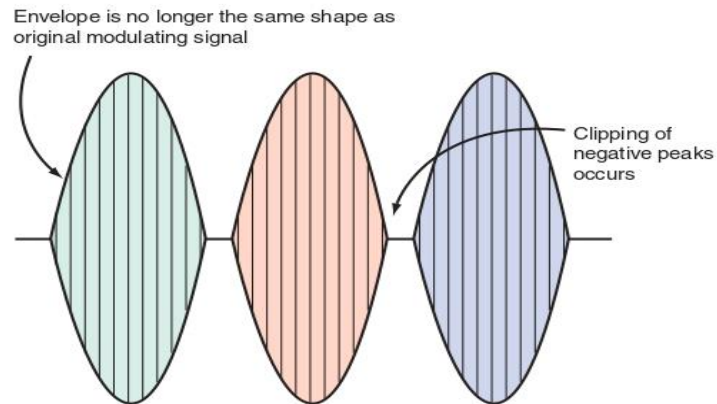


Figure 1.4 Distortion of the envelope caused by overmodulation where the modulating signal amplitude V_m is greater than the carrier signal V_c .

Overmodulation can be prevented by using a circuit called as compression circuit. For example, at different times during voice transmission voices will go from low amplitude to high amplitude. Normally, the amplitude of the modulating signal is adjusted so that only the voice peaks produce 100 percent modulation. This prevents overmodulation and distortion.

Automatic circuits called *compression circuits* solve this problem by amplifying the lower-level signals and suppressing or compressing the higher-level signals. The result is a higher average power output level without overmodulation.

Percentage of Modulation

The modulation index can be determined by measuring the actual values of the modulation voltage and the carrier voltage and computing the ratio.

The modulation index can be computed from $V_{\max}$ and $V_{\min}$, from the AM wave as shown in Fig. 1-5.

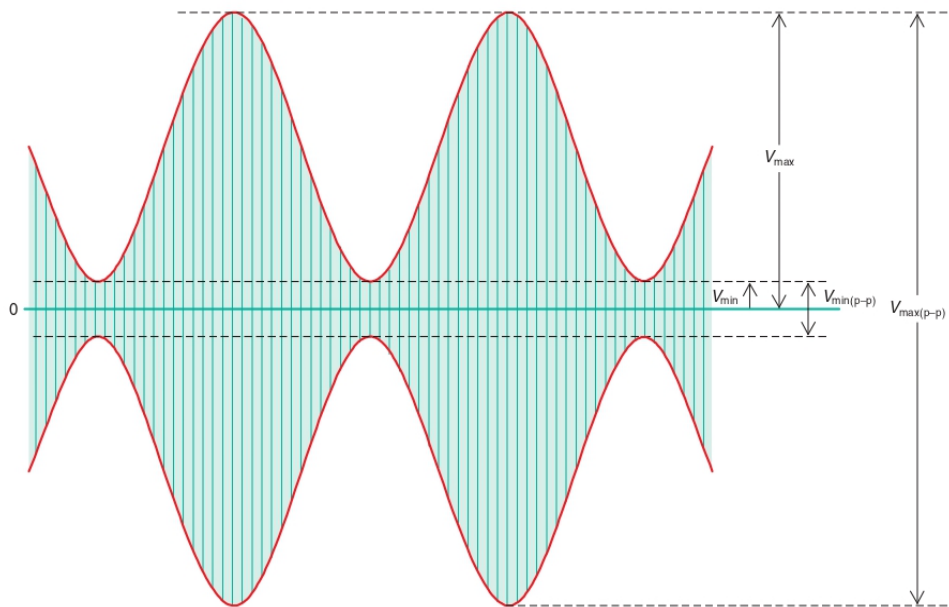


Figure 1.5 An AM wave showing peaks ($V_{\max}$) and troughs ($V_{\min}$).

The peak value of the modulating signal V_m is one-half the difference of the peak and trough values

As shown in Fig. 1-5, $V_{\max}$ is the peak value of the modulated wave, and $V_{\min}$ is the lowest value, or trough, of the modulated wave $V_{\max}$ is nothing but one-half the peak-to-peak value of the AM signal.

Subtracting $V_{\min}$ from $V_{\max}$ produces the peak-to-peak value of the modulating signal. One-half of that, is the peak value of modulating signal

$$V_m = \frac{V_{\max} - V_{\min}}{2} \quad \text{---(1)}$$

The peak value of the carrier signal V_c is the average of the $V_{\max}$ and $V_{\min}$ values:

$$V_c = \frac{V_{\max} + V_{\min}}{2} \quad \text{----- (2)}$$

Modulation index is

$$m = \frac{V_m}{V_c} \quad \text{--(3)}$$

Substituting (1) & (2) in (3)

$$m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$$

The amount, or depth, of AM is more commonly expressed as the percentage of modulation rather than as a fractional value. the percentage of modulation is $100 \times 3\%$

Example 1.1

Suppose that on an AM signal, the $V_{\max(p-p)}$ value read from the graticule on the oscilloscope screen is 5.9 divisions and $V_{\min(p-p)}$ is 1.2 divisions.

a. What is the modulation index?

$$m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} = \frac{5.9 - 1.2}{5.9 + 1.2} = \frac{4.7}{7.1} = 0.662$$

b. Calculate V_c , V_m , and m if the vertical scale is 2 V per division. (*Hint: Sketch the signal.*)

$$V_c = \frac{V_{\max} + V_{\min}}{2} = \frac{5.9 + 1.2}{2} = \frac{7.1}{2} = 3.55 @ \frac{2 \text{ V}}{\text{div}}$$

$$V_c = 3.55 \times 2 \text{ V} = 7.1 \text{ V}$$

$$V_m = \frac{V_{\max} - V_{\min}}{2} = \frac{5.9 - 1.2}{2} = \frac{4.7}{2}$$

$$= 2.35 @ \frac{2 \text{ V}}{\text{div}}$$

$$V_m = 2.35 \times 2 \text{ V} = 4.7 \text{ V}$$

$$m = \frac{V_m}{V_c} = \frac{4.7}{7.1} = 0.662$$

Sidebands and the Frequency Domain

Whenever a carrier is modulated by an information signal, new signals at different frequencies are generated as part of the process. These new frequencies, which are called *side frequencies*, or *sideband*

Sidebands will occur in the frequency spectrum directly above and directly below the carrier frequency. More specifically, the sidebands occur at frequencies that are the sum and difference of the carrier and modulating frequencies. i.e $f_c + f_m$ and $f_c - f_m$ where f_c is carrier frequency and f_m is information signal frequency.

When signals of more than one frequency make up a waveform, it is often better to show the AM signal in the frequency domain rather than in the time domain.

Sideband Calculations

When only a single-frequency sine wave modulating signal is used, the modulation process generates two sidebands. Upper sideband and lower sideband.

If the modulating signal is a complex wave, such as voice or video, a whole range of frequencies modulate the carrier, and thus a whole range of sidebands are generated

The upper sideband f_{USB} and lower sideband f_{LSB} are computed as

$$f_{USB} = f_c + f_m \quad \text{and} \quad f_{LSB} = f_c - f_m$$

where f_c is the carrier frequency and f_m is the modulating frequency.

The existence of sidebands can be demonstrated mathematically, starting with the equation for an AM signal described previously:

$$v_{AM} = V_c \sin 2\pi f_c t + (V_m \sin 2\pi f_m t) (\sin 2\pi f_c t)$$

By using the trigonometric identity that says that the product of two sine waves is

$$\sin A \sin B = \frac{\cos (A - B)}{2} - \frac{\cos (A + B)}{2}$$

and substituting this identity into the expression a modulated wave, the instantaneous amplitude of the signal becomes

$$v_{AM} = V_c \sin 2\pi f_c t + \frac{V_m}{2} \cos 2\pi t(f_c - f_m) - \frac{V_m}{2} \cos 2\pi t(f_c + f_m)$$

where the first term is the carrier; the second term, containing the difference $f_c - f_m$, is the lower sideband; and the third term, containing the sum $f_c + f_m$, is the upper sideband.

For example, assume that a 400-Hz tone modulates a 300-kHz carrier. The upper and lower sidebands are

$$f_{\text{USB}} = 300,000 + 400 = 300,400 \text{ Hz or } 300.4 \text{ kHz}$$

$$f_{\text{LSB}} = 300,000 - 400 = 299,600 \text{ Hz or } 299.6 \text{ kHz}$$

An AM signal is really a composite signal formed from several components: the carrier sine wave is added to the upper and lower sidebands, as the equation indicates. This is illustrated graphically in Fig. 1-6.

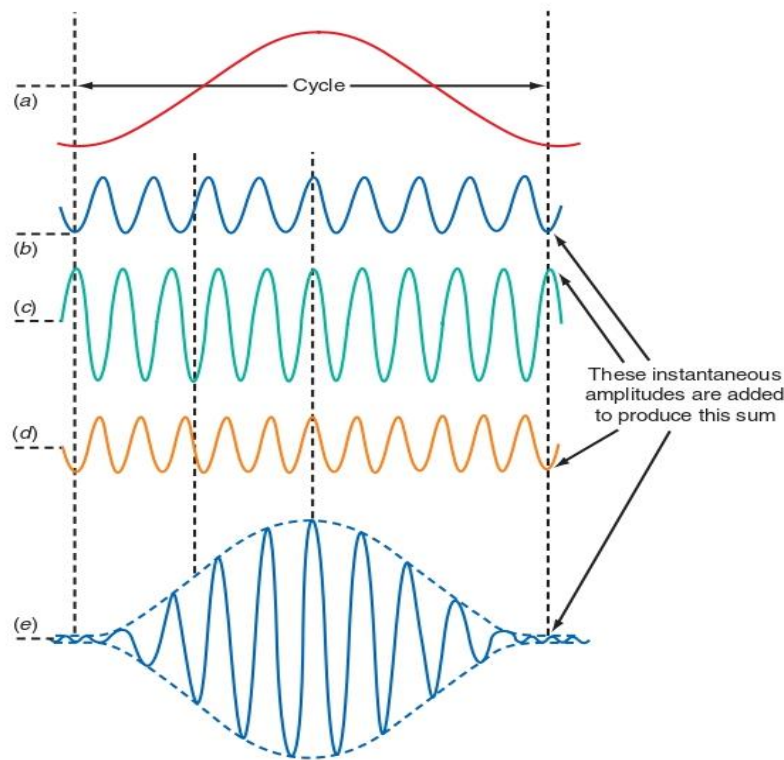


Figure 1.6 The AM wave is the algebraic sum of the carrier and upper and lower sideband sine waves. (a) Intelligence or modulating signal. (b) Lower sideband. (c) Carrier. (d) Upper sideband. (e) Composite AM wave.

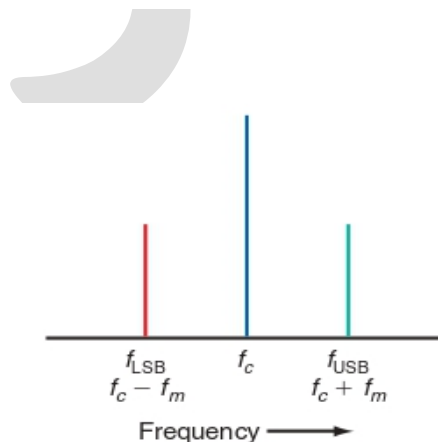


Figure 1.7 A frequency-domain display of an AM signal (voltage).

Adding these signals together algebraically at every instantaneous point along the time axis and plotting the result yield the AM wave shown in the figure. It is a sine wave at the carrier frequency whose amplitude varies as determined by the modulating signal

Frequency-Domain Representation of AM

A plot of signal amplitude versus frequency is referred to as a *frequency-domain display*. A test instrument known as a *spectrum analyzer* is used to display the frequency domain of a signal.

Another method of showing the sideband signals is to plot the carrier and sideband amplitudes with respect to frequency, as in Fig. 1-7. Here the horizontal axis represents frequency, and the vertical axis represents the amplitudes of the signals. The signals may be voltage, current, or power amplitudes and may be given in peak or rms values.

Fig. 1-8 shows the relationship between the time- and frequency-domain displays of an AM signal. The time and frequency axes are perpendicular to each other. The amplitudes shown in the frequency-domain display are the peak values of the carrier and sideband sine waves.

Whenever the modulating signal is more complex than a single sine wave tone, multiple upper and lower sidebands are produced by the AM process.

For example, a voice signal consists of many sine wave components of different frequencies mixed together. Recall that voice frequencies occur in the 300- to 3000-Hz range. Therefore, voice signals produce a range of frequencies above and below the carrier frequency, as shown in Fig. 1-9. These sidebands take up spectrum space. The total bandwidth of an AM signal is calculated by computing the maximum and minimum sideband frequencies. This is done by finding the sum and difference of the carrier frequency and maximum modulating frequency (3000 Hz, or 3 kHz, in Fig. 1-9). For example, if the carrier frequency is 2.8 MHz (2800 kHz), then the maximum and minimum sideband frequencies are

$$f_{\text{USB}} = 2800 + 3 = 2803 \text{ kHz} \quad \text{and} \quad f_{\text{LSB}} = 2800 - 3 = 2797 \text{ kHz}$$

The total bandwidth is simply the difference between the upper and lower sideband frequencies:

$$\text{BW} = f_{\text{USB}} - f_{\text{LSB}} = 2803 - 2797 = 6 \text{ kHz}$$

Bandwidth of an AM signal is twice the highest frequency in the modulating signal: $BW = 2f_m$, where f_m is the maximum modulating frequency

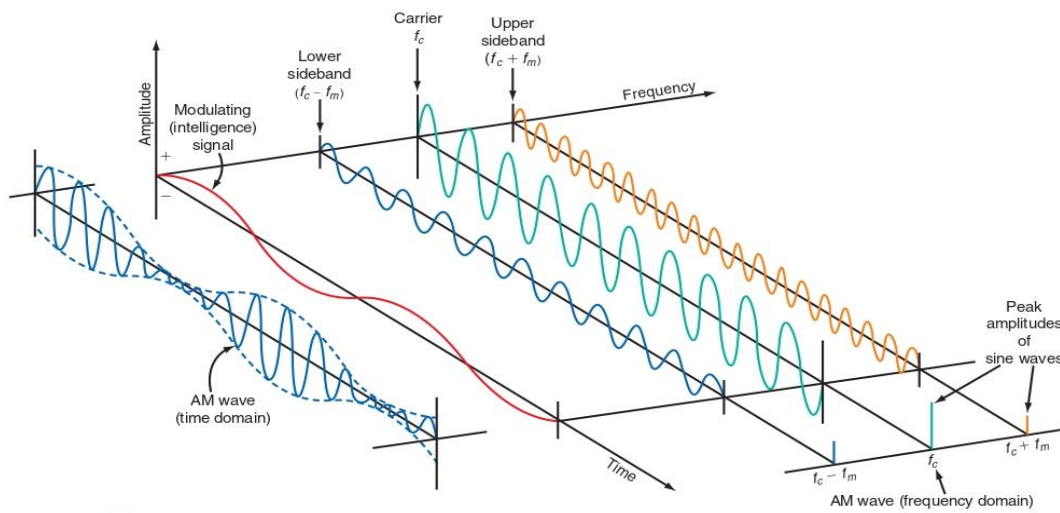


Figure 1.8 The relationship between the time and frequency domains.

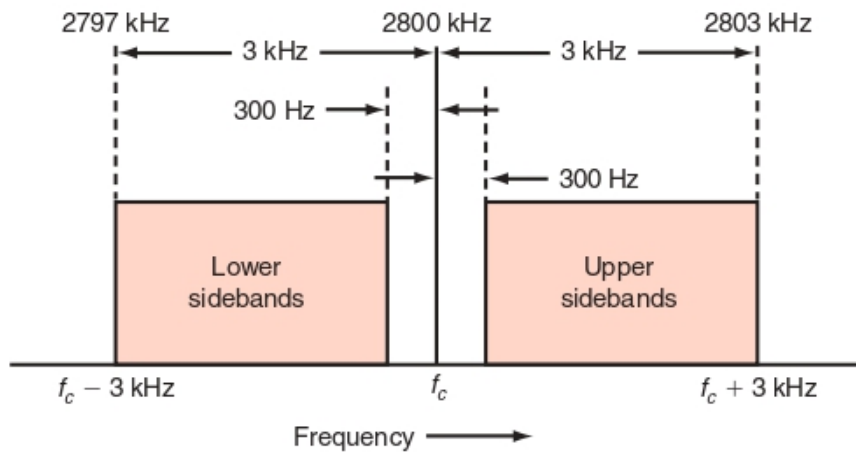


Figure 1.9 The upper and lower sidebands of a voice modulator AM signal.

Example 1.2

A standard AM broadcast station is allowed to transmit modulating frequencies up to 5 kHz. If the AM station is transmitting on a frequency of 980 kHz, compute the maximum and minimum upper and lower sidebands and the total bandwidth occupied by the AM station.

$$f_{\text{USB}} = 980 + 5 = 985 \text{ kHz}$$

$$f_{\text{LSB}} = 980 - 5 = 975 \text{ kHz}$$

$$\text{BW} = f_{\text{USB}} - f_{\text{LSB}} = 985 - 975 = 10 \text{ kHz} \quad \text{or}$$

$$\text{BW} = 2(5 \text{ kHz}) = 10 \text{ kHz}$$

AM Power

In radio transmission, the AM signal is amplified by a power amplifier and fed to the antenna. The AM signal is really a composite of several signal voltages, namely, the carrier and the two sidebands, and each of these signals produces power in the antenna. The total transmitted power P_T is simply the sum of the carrier power P_c and the power in the two sidebands P_{USB} and P_{LSB} :

$$P_T = P_c + P_{\text{LSB}} + P_{\text{USB}}$$

The original AM equation is,

$$v_{\text{AM}} = V_c \sin 2\pi f_c t + \frac{V_m}{2} \cos 2\pi t(f_c - f_m) - \frac{V_m}{2} \cos 2\pi t(f_c + f_m)$$

where the first term is the carrier, the second term is the lower sideband, and the third term is the upper sideband.

Now, remember that V_c and V_m are peak values of the carrier and modulating sine waves, respectively. For power calculations, rms values must be used for the voltages. We can convert from peak to rms by dividing the peak value by $\sqrt{2}$ or multiplying by 0.707. The rms carrier and sideband voltages are then

$$v_{\text{AM}} = \frac{V_c}{\sqrt{2}} \sin 2\pi f_c t + \frac{V_m}{2\sqrt{2}} \cos 2\pi t(f_c - f_m) - \frac{V_m}{2\sqrt{2}} \cos 2\pi t(f_c + f_m)$$

The power in the carrier and sidebands can be calculated by using the power formula $P = V^2/R$, where P is the output power, V is the rms output voltage, and R is the resistive part of the load impedance, which is usually an antenna. We just need to use the coefficients on the sine and cosine terms above in the power formula:

$$P_T = \frac{(V_c/\sqrt{2})^2}{R} + \frac{(V_m/2\sqrt{2})^2}{R} + \frac{(V_m/2\sqrt{2})^2}{R} = \frac{V_c^2}{2R} + \frac{V_m^2}{8R} + \frac{V_m^2}{8R}$$

Remembering that we can express the modulating signal V_m in terms of the carrier V_c by using the expression given earlier for the modulation index $m = V_m/V_c$; we can write

$$V_m = mV_c$$

If we express the sideband powers in terms of the carrier power, the total power becomes

$$P_T = \frac{(V_c)^2}{2R} + \frac{(mV_c)^2}{8R} + \frac{(mV_c)^2}{8R} = \frac{V_c^2}{2R} + \frac{m^2V_c^2}{8R} + \frac{m^2V_c^2}{8R}$$

Since the term $V_c^2/2R$ is equal to the rms carrier power P_c , it can be factored out, giving

$$P_T = \frac{V_c^2}{2R} \left(1 + \frac{m^2}{4} + \frac{m^2}{4} \right)$$

Finally, we get a handy formula for computing the total power in an AM signal when the carrier power and the percentage of modulation are known:

$$P_T = P_c \left(1 + \frac{m^2}{2} \right)$$

For example, if the carrier of an AM transmitter is 1000 W and it is modulated 100 percent ($m = 1$), the total AM power is

$$P_T = 1000 \left(1 + \frac{1^2}{2} \right) = 1500 \text{ W}$$

Of the total power, 1000 W of it is in the carrier. That leaves 500 W in both side-bands. Since the sidebands are equal in size, each sideband has 250 W.

For a 100 percent modulated AM transmitter, the total sideband power is always one-half that of the carrier power

A 50-kW transmitter carrier that is 100 percent modulated will have a sideband power of 25 kW, with 12.5 kW in each sideband. The total power for the AM signal is the sum of the carrier and sideband power, or 75 kW. When the percentage of modulation is less than the optimum 100, there is much less power in the sidebands. For example, for a 70 percent modulated 250-W carrier, the total power in the composite AM signal is

$$P_T = 250 \left(1 + \frac{0.7^2}{2} \right) = 250(1 + 0.245) = 311.25 \text{ W}$$

Of the total, 250 W is in the carrier, leaving $311.25 - 250 = 61.25 \text{ W}$ in the sidebands. There is $61.25/2$ or 30.625 W in each sideband.

Example 1.3

An AM transmitter has a carrier power of 30 W. The percentage of modulation is 85 percent. Calculate (a) the total power and (b) the power in one sideband.

$$\text{a. } P_T = P_c \left(1 + \frac{m^2}{2} \right) = 30 \left[1 + \frac{(0.85)^2}{2} \right] = 30 \left(1 + \frac{0.7225}{2} \right)$$

$$P_T = 30(1.36125) = 40.8 \text{ W}$$

$$\text{b. } P_{\text{SB (both)}} = P_T - P_c = 40.8 - 30 = 10.8 \text{ W}$$

$$P_{\text{SB (one)}} = \frac{P_{\text{SB}}}{2} = \frac{10.8}{2} = 5.4 \text{ W}$$

In the real world, it is difficult to determine AM power by measuring the output voltage and calculating the power with the expression $P = V^2/R$. However, it is easy to measure the current in the load.

the output power is easily calculated by using the formula

$$P_T = I_T^2 R$$

where $I_T = I_c \sqrt{1 + m^2/2}$. Here I_c is the unmodulated carrier current in the load, and m is the modulation index. For example, the total output power of an 85 percent modulated AM transmitter, whose unmodulated carrier current into a $50\text{-}\Omega$ antenna load impedance is 10 A, is

$$I_T = 10\sqrt{1 + \frac{0.85^2}{2}} = 10\sqrt{1.36125} = 11.67 \text{ A}$$

$$P_T = 11.67^2(50) = 136.2(50) = 6809 \text{ W}$$

One way to find the percentage of modulation is to measure both the modulated and the unmodulated antenna currents. Then, by algebraically rearranging the formula above, m can be calculated directly:

$$m = \sqrt{2\left[\left(\frac{I_T}{I_c}\right)^2 - 1\right]}$$

Suppose that the unmodulated antenna current is 2.2 A. That is the current produced by the carrier only, or I_c . Now, if the modulated antenna current is 2.6 A, the modulation index is

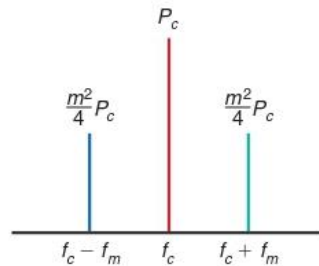
$$m = \sqrt{2\left[\left(\frac{2.6}{2.2}\right)^2 - 1\right]} = \sqrt{2[(1.18)^2 - 1]} = \sqrt{0.7934} = 0.89$$

The percentage of modulation is 89.

As you can see, the power in the sidebands depends on the value of the modulation index. The greater the percentage of modulation, the higher the sideband power and the higher the total power transmitted. Of course, maximum power appears in the sidebands when the carrier is 100 percent modulated. The power in each sideband P_{SB} is given by

$$P_{SB} = P_{LSB} = P_{USB} = \frac{P_c m^2}{4}$$

An example of a time-domain display of an AM signal (power) is as follows.



Example 1.4

An antenna has an impedance of $40\ \Omega$. An unmodulated AM signal produces a current of 4.8 A. The modulation is 90 percent. Calculate (a) the carrier power, (b) the total power, and (c) the sideband power.

a. $P_c = I_c^2 R = (4.8)^2(40) = (23.04)(40) = 921.6\ \text{W}$

b. $I_T = I_c \sqrt{1 + \frac{m^2}{2}} = 4.8 \sqrt{1 + \frac{(0.9)^2}{2}} = 4.8 \sqrt{1 + \frac{0.81}{2}}$
 $I_T = 4.8 \sqrt{1.405} = 5.7\ \text{A}$

$P_T = I_T^2 R = (5.7)^2(40) = 32.49(40) = 1295\ \text{W}$

c. $P_{SB} = P_T - P_c = 1295 - 921.6 = 373.4\ \text{W}$ (186.7 W each sideband)

Example 1.5

The transmitter in Example 1.4 experiences an antenna current change from 4.8 A unmodulated to 5.1 A. What is the percentage of modulation?

$$\begin{aligned} m &= \sqrt{2 \left[\left(\frac{I_T}{I_c} \right)^2 - 1 \right]} \\ &= \sqrt{2 \left[\left(\frac{5.1}{4.8} \right)^2 - 1 \right]} \\ &= \sqrt{2[(1.0625)^2 - 1]} \\ &= \sqrt{2(1.13 - 1)} \\ &= \sqrt{2(0.13)} \\ &= \sqrt{0.26} \\ m &= 0.51 \end{aligned}$$

The percentage of modulation is 51.

Example 1.6

What is the power in one sideband of the transmitter in Example 1.4

$$P_{SB} = m^2 \frac{P_c}{4} = \frac{(0.9)^2(921.6)}{4} = \frac{746.5}{4} = 186.6\ \text{W}$$

Single-Sideband Modulation

In amplitude modulation, two-thirds of the transmitted power is in the carrier, which itself conveys no information. The real information is contained within the sidebands. One way to improve the efficiency of amplitude modulation is to suppress the carrier and eliminate one sideband. The result is a single-sideband (SSB) signal. SSB is a form of AM that offers unique benefits in some types of electronic communication.

DSB Signals

The first step in generating an SSB signal is to suppress the carrier, leaving the upper and lower sidebands. This type of signal is referred to as a *double-sideband suppressed carrier (DSSC or DSB)* signal. The benefit, of course, is that no power is wasted on the carrier. Double-sideband suppressed carrier modulation is simply a special case of AM with no carrier.

A typical DSB signal is shown in Fig. 1-10. This signal, the algebraic sum of the two sinusoidal sidebands, is the signal produced when a carrier is modulated by a single-tone sine wave information signal. The carrier is suppressed

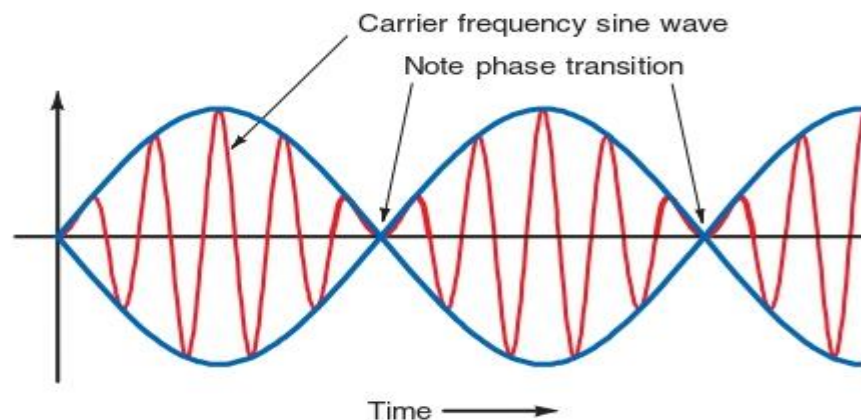
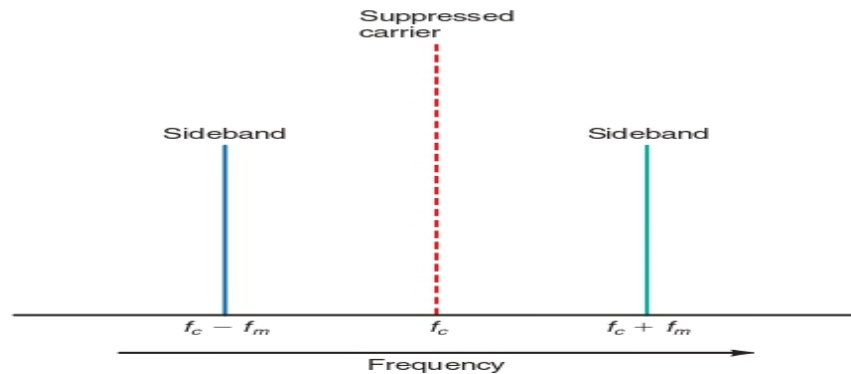


Figure 1.10 A time-domain display of a DSB AM signal.

Figure 1.11 A frequency-domain display of DSB signal.

In Fig. 1-10, note that there are two adjacent positive-going half-cycles at the null points in the wave. That is one way to tell from an oscilloscope display whether the signal shown is a true DSB signal.

A *frequency-domain display* of a DSB signal is given in Fig. 1-11. As shown, the spectrum space occupied by a DSB signal is the same as that for a conventional AM signal.

Double-sideband suppressed carrier signals are generated by a circuit called a *balanced modulator*. The purpose of the balanced modulator is to produce the sum and difference frequencies but to cancel or balance out the carrier.

SSB Signals

In DSB transmission, since the sidebands are the sum and difference of the carrier and modulating signals, the information is contained in both sidebands. As it turns out, there is no reason to transmit both sidebands in order to convey the information. One sideband can be suppressed; the remaining sideband is called a *single-sideband suppressed carrier* (SSSC or SSB) signal. SSB signals offer four major benefits

1. The primary benefit of an SSB signal is that the spectrum space it occupies is only one-half that of AM and DSB signals. This greatly conserves spectrum space and allows more signals to be transmitted in the same frequency range.
2. All the power previously devoted to the carrier and the other sideband can be channeled into the single sideband, producing a stronger signal that should carry farther and be more reliably

received at greater distances. Alternatively, SSB transmitters can be made smaller and lighter than an equivalent AM or DSB transmitter because less circuitry and power are used

3. Because SSB signals occupy a narrower bandwidth, the amount of noise in the signal is reduced

4. There is less selective fading of an SSB signal over long distances. An AM signal is really multiple signals, at least a carrier and two sidebands. These are on different frequencies, so they are affected in slightly different ways by the ionosphere and upper atmosphere, which have a great influence on radio signals of less than about 50 MHz. The carrier and sidebands may arrive at the receiver at slightly different times, causing a phase shift that can, in turn, cause them to add in such a way as to cancel one another rather than add up to the original AM signal. Such cancellation, or *selective fading*, is not a problem with SSB since only one sideband is being transmitted.

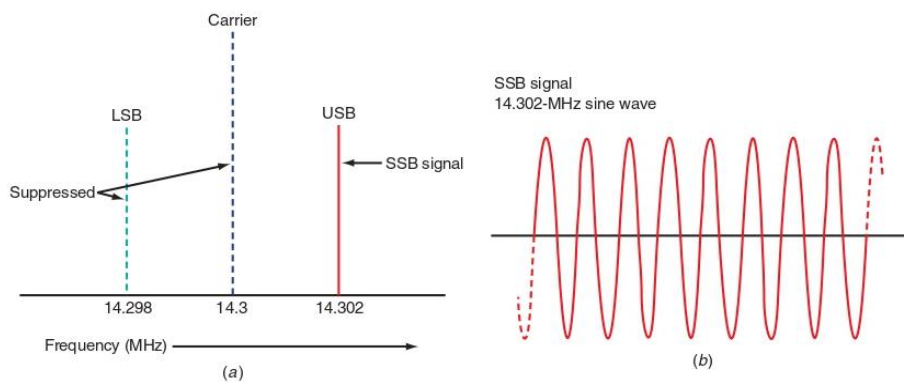


Figure 1.12 An SSB signal produced by a 2-kHz sine wave modulating a 14.3-MHz sine wave carrier.

Fig. 1-12 shows the frequency- and time-domain displays of an SSB signal produced when a steady

2-kHz sine wave tone modulates a 14.3-MHz carrier. Amplitude modulation would produce sidebands of 14.298 and 14.302 MHz. In SSB, only one sideband is used. Fig. 1-12(a) shows that only the upper sideband is generated. The RF signal is simply a constant-power 14.302-MHz sine wave. A time-domain display of this SSB signal is shown in Fig. 1-12(b).

Of course, most information signals transmitted by SSB are not pure sine waves. A more common modulation signal is voice, with its varying frequency and amplitude content. The voice signal creates a complex RF SSB signal that varies in frequency

and amplitude over the narrow spectrum defined by the voice signal bandwidth. The wave- form at the output of the SSB modulator has the same shape as the baseband waveform, but it is shifted in frequency.

Disadvantages of DSB and SSB

The main disadvantage of DSB and SSB signals is that they are harder to recover, or demodulate, at the receiver.

Demodulation depends upon the carrier being present. If the carrier is not present, then it must be regenerated at the receiver and reinserted into the signal. To faithfully recover the information signal, the reinserted carrier must have the same phase and frequency as those of the original carrier. This is a difficult requirement.

To solve this problem, a low-level carrier signal is sometimes transmitted along with the two sidebands in DSB or a single sideband in SSB. Because the carrier has a low power level, the essential benefits of SSB are retained, but a weak carrier is received so that it can be amplified and reinserted to recover the original information. Such a low-level carrier is referred to as a *pilot carrier*. This technique is used in FM stereo transmissions as well as in the transmission of the color information in a TV picture.

Signal power considerations

In SSB, the transmitter output is expressed in terms of *peak envelope power (PEP)*, the maximum power produced on voice amplitude peaks. PEP is computed by the equation $P = V^2/R$. For example, assume that a voice signal produces a 360-V, peak-to-peak signal across a 50- Ω load. The rms voltage is 0.707 times the peak value, and the peak value is one-half the peak-to-peak voltage. In this example, the rms voltage is $0.707(360/2) = 127.26$ V.

The peak envelope power is then

$$\text{PEP} = V_{\text{rms}}^2/R = \frac{(127.26)^2}{50} = 324 \text{ W}$$

The PEP input power is simply the dc input power of the transmitter's final amplifier stage at the instant of the voice envelope peak. It is the final amplifier stage dc supply voltage multiplied by the maximum amplifier current that occurs at the peak, or

$$\text{PEP} = V_s I_{\text{max}}$$

where V_s = amplifier supply voltage

I_{max} = current peak

For example, a 450-V supply with a peak current of 0.8 A produces a PEP of $450(0.8) = 360$ W.

Note that voice amplitude peaks are produced only when very loud sounds are generated during certain speech patterns or when some word or sound is emphasized. During normal speech levels, the input and output power levels are much less than the PEP level. The average power is typically only one-fourth to one-third of the PEP value with typical human speech:

$$P_{\text{avg}} = \frac{\text{PEP}}{3} \quad \text{or} \quad P_{\text{avg}} = \frac{\text{PEP}}{4}$$

With a PEP of 240 W, the average power is only 60 to 80 W. Typical SSB transmitters are designed to handle only the average power level on a continuous basis, not the PEP.

The transmitted sideband will, of course, change in frequency and amplitude as a complex voice signal is applied. This sideband will occupy the same bandwidth as one sideband in a fully modulated AM signal with carrier.

Example 1.7

An SSB transmitter produces a peak-to-peak voltage of 178 V across a 75-Ω antenna load. What is the PEP?

$$V_p = \frac{V_{p-p}}{2} = \frac{178}{2} = 89 \text{ V}$$

$$V_{\text{rms}} = 0.707 V_p = 0.707(89) = 62.9 \text{ V}$$

$$P = \frac{V^2}{R} = \frac{(62.9)^2}{75} = 52.8 \text{ W}$$

$$\text{PEP} = 52.8 \text{ W}$$

Example 1.8

An SSB transmitter has a 24-V dc power supply. On voice peaks the current achieves a maximum of 9.3 A.

a. What is the PEP?

$$\text{PEP} = V_s I_m = 24(9.3) = 223.2 \text{ W}$$

b. What is the average power of the transmitter?

$$P_{\text{avg}} = \frac{\text{PEP}}{3} = \frac{223.2}{3} = 74.4 \text{ W}$$

$$P_{\text{avg}} = \frac{\text{PEP}}{4} = \frac{223.2}{4} = 55.8 \text{ W}$$

$$P_{\text{avg}} = 55.8 \text{ to } 74.4 \text{ W}$$

Basic Principles of Amplitude modulation

Basic equation of AM signal is

$$v_{\text{AM}} = V_c \sin 2\pi f_c t + (V_m \sin 2\pi f_m t)(\sin 2\pi f_c t)$$

where the first term is the sine wave carrier and second term is the product of the sinewave carrier and modulating signals. (v_{AM} is the instantaneous value of the amplitude modulation voltage.)

The modulation index m is the ratio of the modulating signal amplitude to the carrier amplitude, or $m = V_m/V_c$, and so $V_m = mV_c$. Then substituting this for V_m in the basic equation gives

$$v_{\text{AM}} = V_c \sin 2\pi f_c t + (mV_c \sin 2\pi f_m t)(\sin 2\pi f_c t)$$

$$\text{Factoring gives } v_{\text{AM}} = V_c \sin 2\pi f_c t (1 + m \sin 2\pi f_m t).$$

AM in the Time Domain

From the expression for v_{AM} , it is clear that we need a circuit that can multiply the

carrier by the modulating signal and then add the carrier. A block diagram of such a circuit is shown in Fig. 1-13

One way to do this is to develop a circuit whose gain (or attenuation) is a function of $1 + m \sin 2\pi f_m t$. If we call that gain A , the expression for the AM signal becomes

$$v_{AM} = A(v_c)$$

where A is the gain or attenuation factor

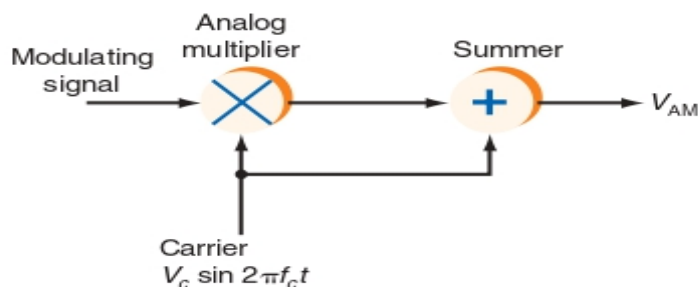


Figure 1.13 Block diagram of a circuit to produce AM.

where A is the gain or attenuation factor. Fig. 1-14 shows simple circuits based on this expression. In Fig. 1-14(a), A is a gain greater than 1 provided by an amplifier. In Fig. 1-14(b), the carrier is attenuated by a voltage divider. The gain in this case is less than 1 and is therefore an attenuation factor. The carrier is multiplied by a fixed fraction A .

Now, if the gain of the amplifier or the attenuation of the voltage divider can be varied in accordance with the modulating signal plus 1, AM will be produced.

In Fig. 1-14(a) the modulating signal would be used to increase or decrease the gain of the amplifier as the amplitude of the intelligence changed. In Fig. 1-14(b), the modulating signal could be made to vary one of the resistances in the voltage divider, creating a varying attenuation factor.

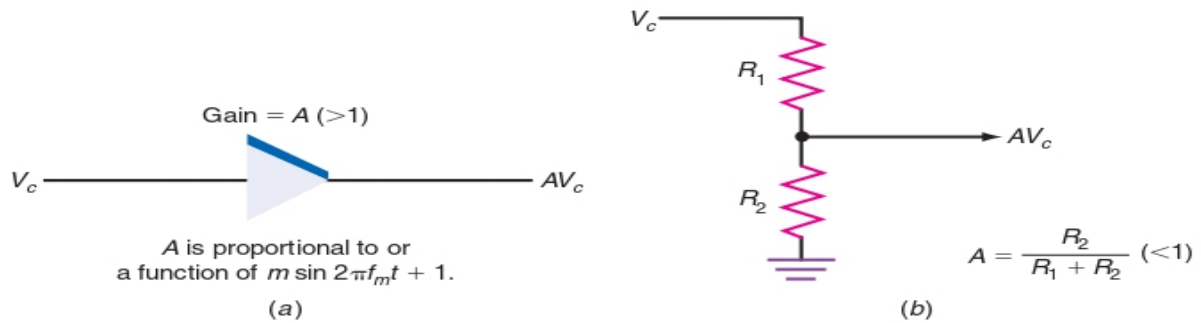


Figure 1.14 Multiplying the carrier by a fixed gain A .

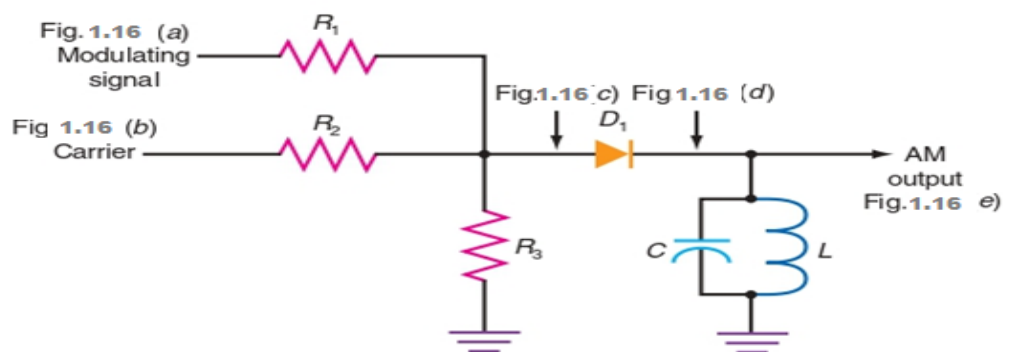
Amplitude Modulators

Amplitude modulators are generally one of two types: low level or high level. Low-level modulators generate AM with small signals and thus must be amplified considerably if they are to be transmitted. High-level modulators produce AM at high power levels, usually in the final amplifier stage of a transmitter.

Low-Level AM Diode Modulator

One of the simplest amplitude modulators is the *diode modulator*. The practical implementation shown in Fig. 1-15 consists of a resistive mixing network, a diode rectifier, and an LC tuned circuit

Figure 1.15 Amplitude modulation with a diode.

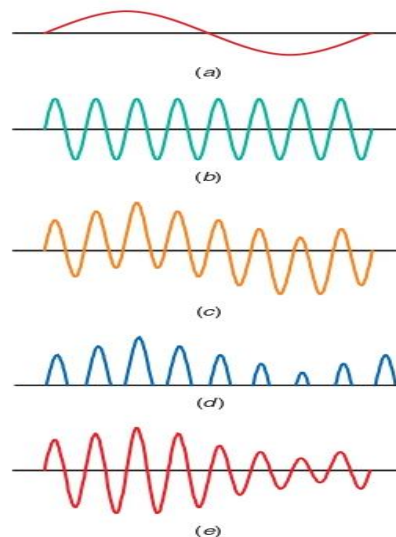


The carrier [Fig. 1.16(b)] is applied to one input resistor and the modulating signal [1.16(a)] to the other. The mixed signals appear across R_3 . This network causes the two signals to be linearly mixed, i.e., algebraically added. If both the carrier and the modulating signal are sine waves,

the waveform resulting at the junction of the two resistors will be like that shown in [Fig. 1.16(c)]. This signal is not AM. Modulation is a multiplication process, not an addition process. The composite waveform is applied to a diode rectifier. The diode is connected so that it is forward-biased by the positive-going half-cycles of the input wave. During the negative portions of the wave, the diode is cut off and no signal passes. The current through the diode is a series of positive-going pulses whose amplitude varies in proportion to the amplitude of the modulating signal [Fig. 1-16(d)].

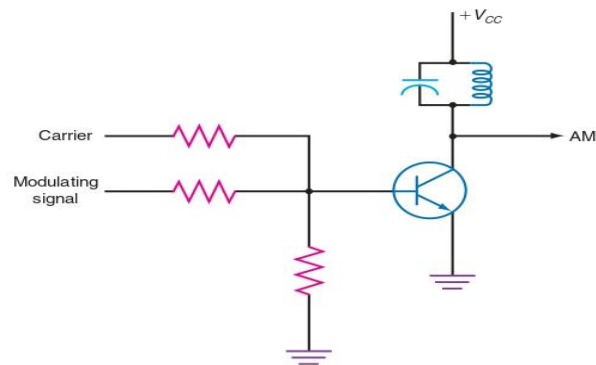
These positive-going pulses are applied to the parallel-tuned circuit made up of L and C , which are resonant at the carrier frequency. Each time the diode conducts, a pulse of current flows through the tuned circuit. The coil and capacitor repeatedly exchange energy, causing an oscillation, or “ringing,” at the resonant frequency. The oscillation of the tuned circuit creates one negative half-cycle for every positive input pulse. High- amplitude positive pulses cause the tuned circuit to produce high-amplitude negative pulses. Low-amplitude positive pulses produce corresponding low-amplitude negative pulses. The resulting waveform across the tuned circuit is an AM signal, as Fig. 1-16(e) illustrates.

Figure 1.16 Waveforms in the diode modulator. (a) Modulating signal. (b) Carrier. (c) Linearly mixed modulating signal and carrier. (d) Positive-going signal after diode D_1 . (e) Am output signal.



Transistor Modulator

Figure 1.17 Simple transistor modulator.



It uses a transistor instead of the diode. The emitter-base junction is a diode and a nonlinear device. The composite waveform is applied to base of transistor. Rectification occurs because of the emitter-base junction which is a diode. The emitter base junction is forward-biased by the positive-going half-cycles of the input wave. During the negative portions of the wave, it is cut off and no signal passes.

This causes larger half-sine pulses of current in the tuned circuit. The tuned circuit oscillates (rings) to generate the missing half-cycle. The output is a classic AM wave.

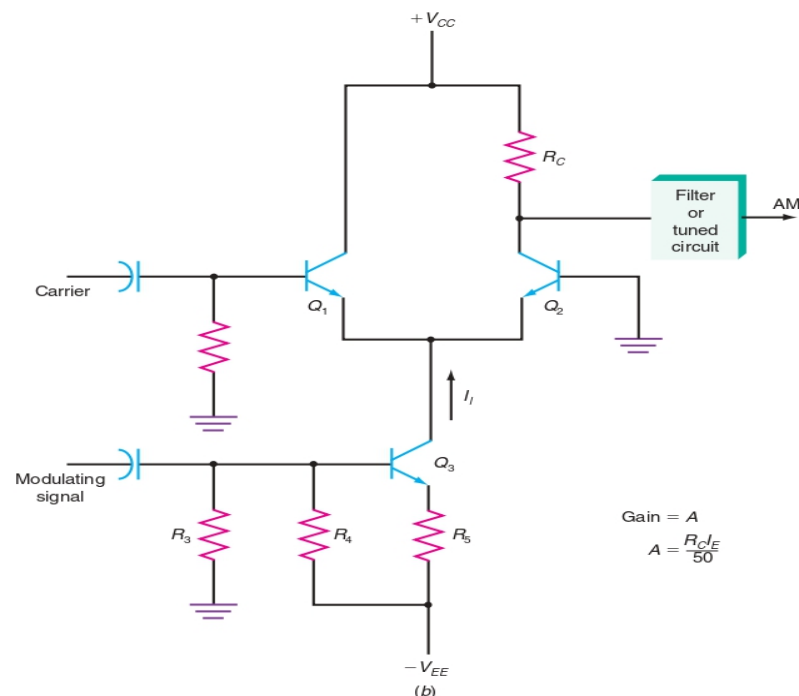
Differential Amplifier Modulator

A differential amplifier modulator makes an excellent amplitude modulator. Transistors Q_1 and Q_2 form the differential pair, and Q_3 is a constant-current source. Transistor Q_3 supplies a fixed emitter current I_E to Q_1 and Q_2 , one-half of which flows in each transistor. The output is developed across the collector resistors R_1 and R_2 .

The carrier is applied to the base of Q_1 , and the base of Q_2 is grounded. The output, taken from the collector of Q_2 , is single-ended. Since the output from Q_1 is not used, its collector resistor can be omitted with no effect on the circuit. The modulating signal is applied to the base of the constant-

current source Q_3 . As the intelligence signal varies, it varies the emitter current. This changes the gain of the circuit, amplifying the carrier by an amount determined by the modulating signal amplitude. The result is AM in the output

The differential amplifier makes an excellent amplitude modulator. It has a high gain and good linearity, and it can be modulated 100 percent. And if high-frequency transistors or a high-frequency IC differential amplifier is used, this circuit can be used to produce low-level modulation at frequencies well into the hundreds of megahertz.



High-Level AM

In high-level AM, the modulator varies the voltage and power in the final RF amplifier stage of the transmitter. The result is high efficiency in the RF amplifier and overall high-quality performance.

Collector Modulator

One example of a high-level modulator circuit is the *collectormodulator* shown in Fig. 1-19.

The modulating signal is applied to the collector of the transistor and hence this circuit is called collector modulator .

The output stage of the transmitter is a high-power class C amplifier. Class C amplifiers conduct for only a portion of the positive half-cycle of their input signal. Class C amplifiers are used for radio applications as their efficiency and distortion handling capability is very good. Class C amplifiers conduct for less than 180 degree.

The collector current pulses cause the tuned circuit to oscillate at the desired output frequency.

Figure 1.19 Low-level modulation systems use linear power amplifiers to increase the AM signal level before transmission.

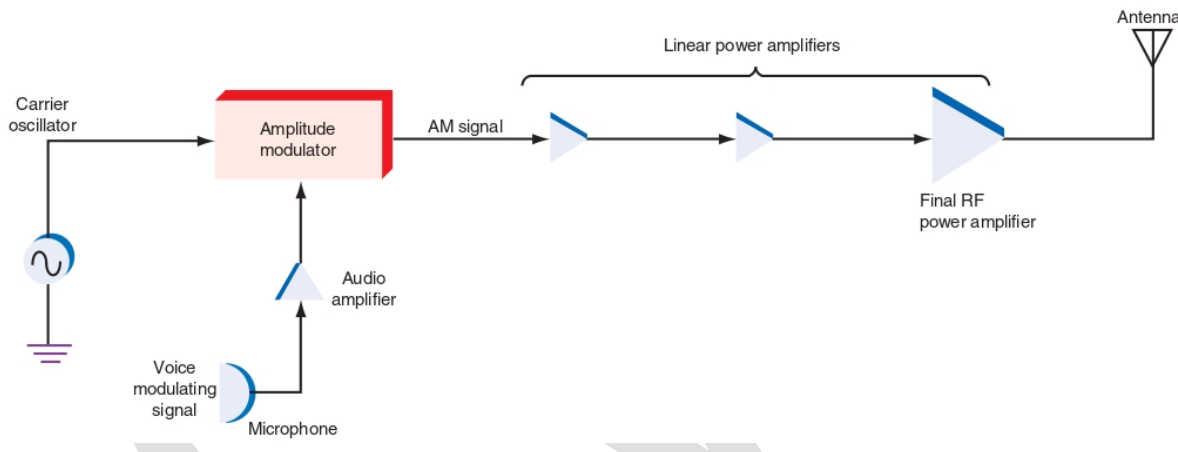
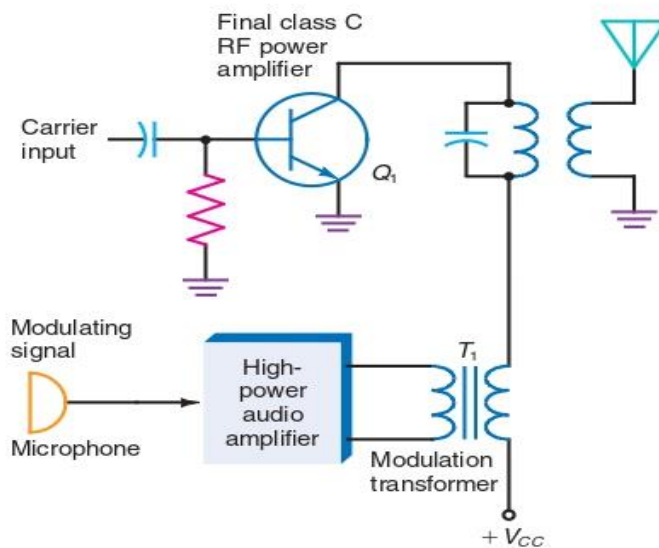


Figure 1.20 A high-level collector modulator.



The modulator is a linear power amplifier that takes the low-level modulating signal

and amplifies it to a high-power level. The modulating output signal is coupled through modulation transformer T_1 to the class C amplifier.

The secondary winding of the modulation transformer is connected in series with the collector supply voltage V_{CC} of the class C amplifier. With a zero-modulation input signal, there is zero-modulation voltage across the secondary of T_1 , the collector supply voltage is applied directly to the class C amplifier, and the output carrier is a steady sine wave.

When the modulating signal occurs, the ac voltage of the modulating signal across the secondary of the modulation transformer is added to and subtracted from the dc collector supply voltage. This varying supply voltage is then applied to the class C amplifier, causing the amplitude of the current pulses through transistor Q_1 to vary. As a result, the amplitude of the carrier sine wave varies in accordance with the modulated signal. When the modulation signal goes positive, it adds to the collector supply voltage, thereby increasing its value and causing higher current pulses and a higher-amplitude carrier. When the modulating signal goes negative, it subtracts from the collector supply voltage, decreasing it. For that reason, the class C amplifier current pulses are smaller, resulting in a lower-amplitude carrier output.

Example 1.9

An AM transmitter uses high-level modulation of the final RF power amplifier, which has a dc supply voltage V_{CC} of 48 V with a total current I of 3.5 A. The efficiency is 70 percent.

- a. What is the RF input power to the final stage?

$$\text{DC input power} = P_i = V_{CC}I = 48 \times 3.5 = 168 \text{ W}$$

- b. How much AF power is required for 100 percent modulation? (*Hint: For 100 percent modulation, AF modulating power P_m is one-half the input power.*)

$$P_m = \frac{P_i}{2} = \frac{168}{2} = 84 \text{ W}$$

- c. What is the carrier output power?

$$\% \text{ efficiency} = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100$$

$$P_{\text{out}} = \frac{\% \text{ efficiency} \times P_{\text{in}}}{100} = \frac{70(168)}{100} = 117.6 \text{ W}$$

- d. What is the power in one sideband for 67 percent modulation?

$$P_s = \text{sideband power}$$

$$P_s = \frac{P_c(m^2)}{4}$$

$$m = \text{modulation percentage (\%)} = 0.67$$

$$P_c = 168$$

$$P_s = \frac{168(0.67)^2}{4} = 18.85 \text{ W}$$

- e. What is the maximum and minimum dc supply voltage swing with 100 percent modulation? (See Fig. 4-13.)

$$\text{Minimum swing} = 0$$

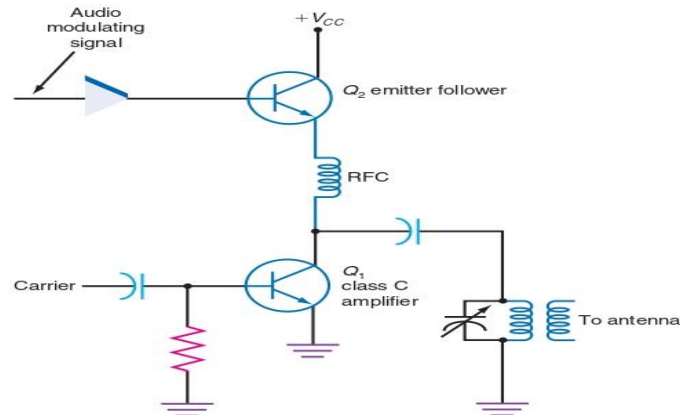
$$\text{Supply voltage } V_{CC} = 48 \text{ V}$$

$$\text{Maximum swing } 2 \times V_{CC} = 2 \times 48 = 96 \text{ V}$$

Series Modulator

A major disadvantage of collector modulators is the need for a modulation transformer that connects the audio amplifier to the class C amplifier in the transmitter. The higher the power, the larger and more expensive the transformer

Figure 1.21 Series modulation. Transistors may also be MOSFETs with appropriate biasing.



This series modulator replaces the transformer with an emitter follower. The modulating signal is applied to the emitter follower Q_2 , which is an audio power amplifier. The emitter follower appears in series with the collector supply voltage V_{CC} . This causes the amplified audio modulating signal to vary the collector supply voltage to the class C amplifier Q_1 , as illustrated in Fig. 1-21.

If the modulating signal goes positive, the supply voltage to Q_1 increases; thus, the carrier amplitude increases in proportion to the modulating signal.

If the modulating signal goes negative, the supply voltage to Q_1 decreases, thereby decreasing the carrier amplitude in proportion to the modulating signal. For 100 percent modulation, the emitter follower can reduce the supply voltage to zero on maximum negative peaks.

Using this high-level modulating scheme eliminates the need for a large, heavy, and expensive transformer, and considerably improves frequency response. However, it is very inefficient. The emitter-follower modulator must dissipate as much power as the class C RF amplifier. For example, assume a collector supply voltage of 24 V and a collector current of 0.5 A. With no modulating signal applied, the percentage of modulation is 0. The emitter follower is biased so that the base and the emitter are at a dc voltage of about one-half the supply voltage, or in this example 12 V. The collector supply voltage on the class C amplifier is 12 V, and the input power is therefore

$$P_{in} = V_{CC}I_c = 12(0.5) = 6 \text{ W}$$

To produce 100 percent modulation, the collector voltage on Q_1 must double, as must the collector current. This occurs on positive peaks of the audio input, as described above. At this time most of the audio signal appears at the emitter of Q_1 ; very little of the signal appears between the emitter and collector of Q_2 , and so at 100 percent modulation, Q_2 dissipates very little power.

When the audio input is at its negative peak, the voltage at the emitter of Q_2 is reduced to 12 V. This means that the rest of the supply voltage, or another 12 V, appears between the emitter and collector of Q_2 . Since Q_2 must also be able to dissipate 6 W, it has to be a very large power transistor. The efficiency drops to less than 50 percent. With a modulation transformer, the efficiency is much greater, in some cases as high as 80 percent.

This arrangement is not practical for very high power AM, but it does make an effective higher-level modulator for power levels below about 100 W.

Amplitude demodulators

Demodulators, or *detectors*, are circuits that accept modulated signals and recover the original modulating information. The demodulator circuit is the key circuit in any radio receiver

Diode Detectors

The simplest and most widely used amplitude demodulator is the *diode detector* (see Fig. 1-22).

The AM signal is usually transformer-coupled and applied to a basic half wave rectifier circuit consisting of D_1 and R_1 . The diode conducts when the positive half-cycles of the AM signals occur. During the negative half-cycles, the diode is reverse-biased and no current flows through it. As a result, the voltage across R_1 is a series of positive pulses whose amplitude varies with the modulating signal. A capacitor C_1 is connected across resistor R_1 , effectively filtering out the carrier and thus recovering the original modulating signal.

Working :

The waveforms in Fig. 1-23 illustrate this.

On each positive half cycle of the AM signal, the capacitor charges quickly to the peak value of the pulses passed by the diode. When the pulse voltage drops to zero, the capacitor discharges into resistor R_1 . The time constant of C_1 and R_1 is chosen to be long compared to the period of the carrier. As a result, the capacitor discharges only slightly during the time that the diode is not

conducting.

When the next pulse comes along, the capacitor again charges to its peak value. When the diode cuts off, the capacitor again discharges a small amount into the resistor. The resulting waveform across the capacitor is a close approximation to the original modulating signal.

Because the capacitor charges and discharges, the recovered signal has a small amount of ripple on it, causing distortion of the modulating signal. However, because the carrier frequency is usually many times higher than the modulating frequency, these ripple variations are barely noticeable.

Because the diode detector recovers the envelope of the AM signal, which is the original modulating signal, the circuit is sometimes referred to as an *envelope detector*.

Distortion of the original signal can occur if the time constant of the load resistor R_1 and the shunt filter capacitor C_1 is too long or too short. If the time constant is too long, the capacitor discharge will be too slow to follow the faster changes in the modulating signal. This is referred to as *diagonal distortion*. If the time constant is too short, the capacitor will discharge too fast and the carrier will not be sufficiently filtered out.

Figure 1.22 A diode detector AM demodulator.

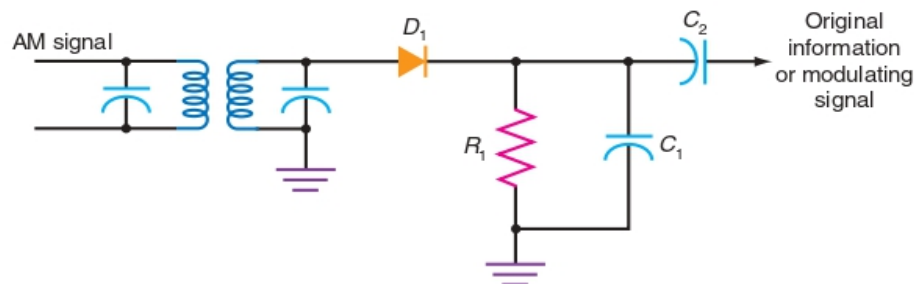
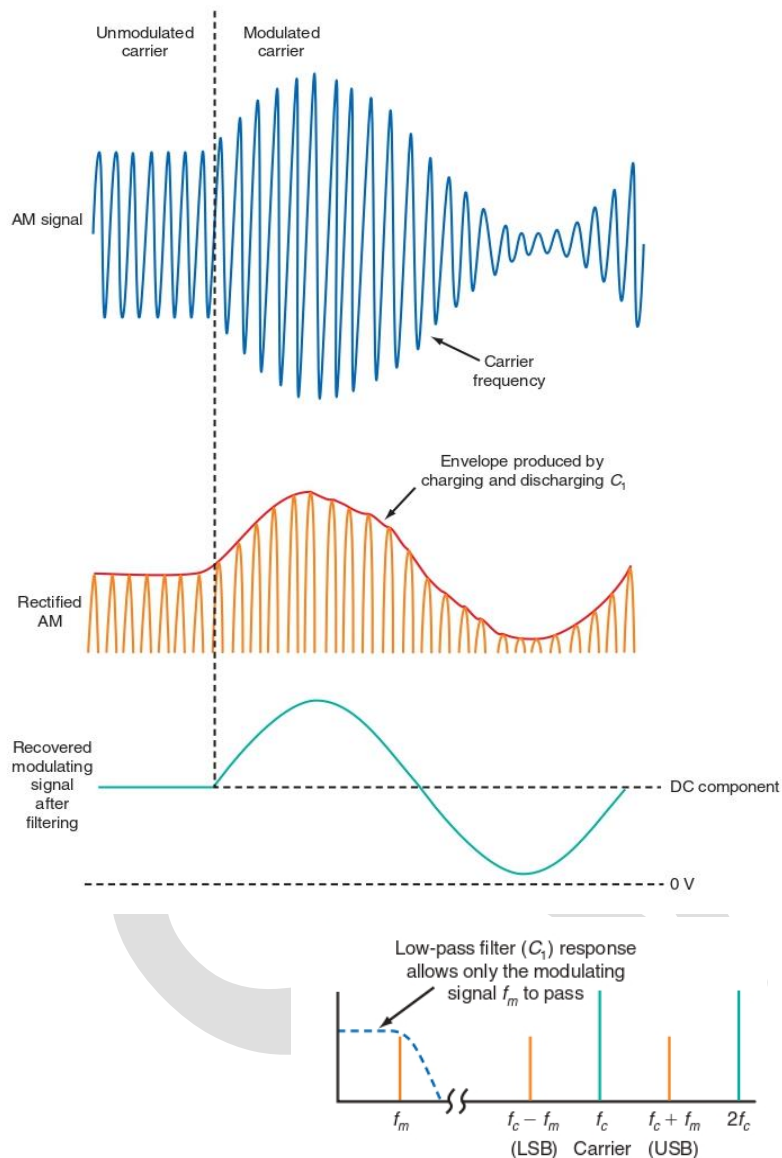


Figure 1.23 Diode detector waveforms.**Figure 1.24** Output spectrum of a diode detector.

In frequency domain the diode is regarded as a nonlinear device to which are applied multiple signals where modulation will take place. The multiple signals are the carrier and sidebands, which make up the input AM signal to be demodulated. The components of the AM signal are the carrier f_c , the upper sideband $f_c + f_m$, and the lower sideband $f_c - f_m$.

Crystal Radio receivers:

The crystal component of the *crystal radio receiver* is simply a diode. A long wire antenna picks up the radio signal, which is inductively coupled to the secondary winding of T_1 , which forms a series resonant circuit with C_1 . Here voltage induced into the secondary winding appears as a voltage source in series with the coil and capacitor. The variable capacitor C_1 is used to select a

station. At resonance, the voltage across the capacitor is stepped up by a factor equal to the Q of the tuned

circuit. This resonant voltage rise is a form of amplification.

This higher-voltage signal is applied to the diode. The diode detector D_1 and its filter C_2 recover the original modulating information, which causes current flow in the headphones. The headphones serve as the load resistance, and capacitor C_2 removes the carrier.

Reception is very weak because no active amplification is provided.

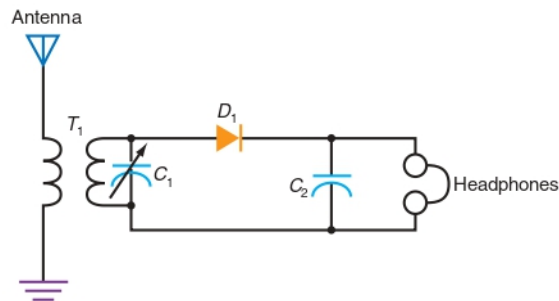


Figure 1.25 A crystal radio receiver.

Synchronous detection:

Synchronous detectors use an internal clock signal at the carrier frequency in the receiver to switch the AM signal off and on, producing rectification similar to that in a standard diode detector as shown in figure 1.26.

The AM signal is applied to a series switch that is opened and closed synchronously with the carrier signal. The switch is usually a diode or transistor that is turned on or off by an internally generated clock signal equal in frequency to and in phase with the carrier frequency.

The switch in Fig. 1.26 is turned on by the clock signal during the positive half-cycles of the AM signal, which therefore appears across the load resistor.

During the negative half-cycles of the AM signal, the clock turns the switch off, so no signal reaches the load or filter capacitor. The capacitor filters out the carrier.

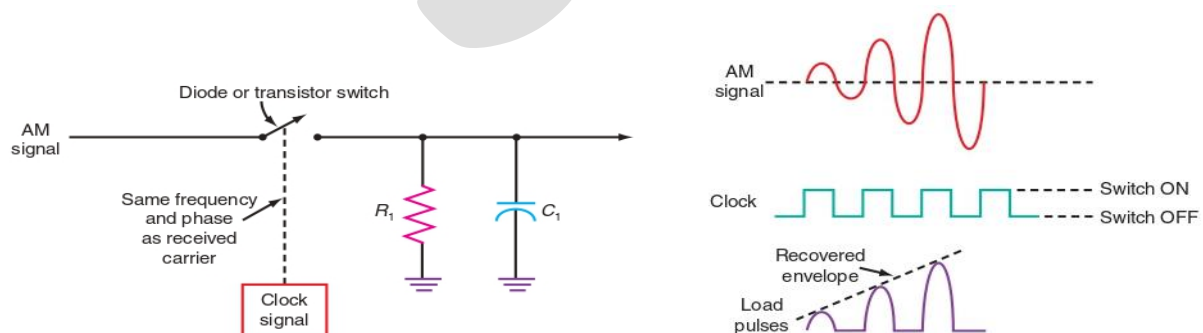
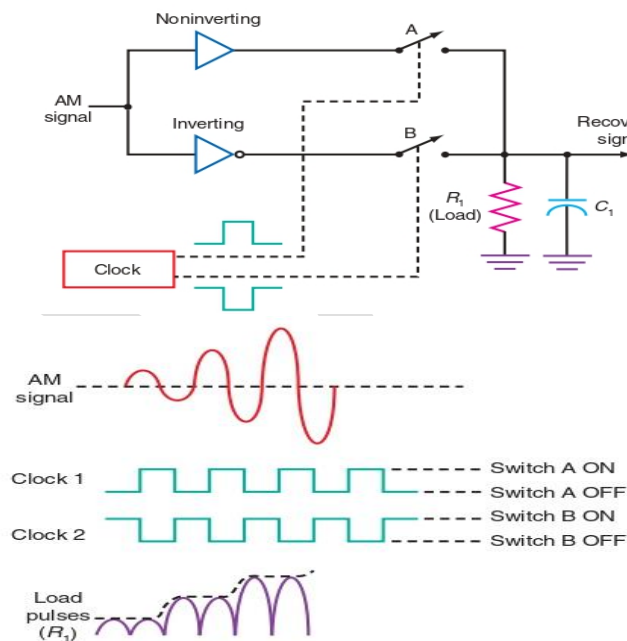


Figure 1.26 Concept of a synchronous detector, and waveform

A **full wave synchronous detector** is shown in Fig. 1.27.

The AM signal is applied to both inverting and noninverting amplifiers. The internally generated carrier signal operates two switches A and B.

During positive half-cycles of the AM signal, the switch A feeds the noninverted AM output of positive half-cycles to the load. During the negative half-cycles of the input, the switch B connects the output of the inverter to the load. The negative half-cycles are inverted, becoming positive, and the signal appears across the load. The result is full wave rectification of the signal.

**Figure 1.27** A full-wave synchronous detector.

Carrier recovery circuit:

**Figure 1.28** A simple carrier recovery circuit.

The circuit shown in Fig. 1-28 is one way to supply the carrier to the synchronous

detector.

The AM signal to be demodulated is applied to a highly selective bandpass filter that picks out the carrier and suppresses the sidebands, thus removing most of the amplitude variations. This signal is amplified and applied to a clipper or limiter that removes any remaining amplitude variations from the signal, leaving only the carrier. The clipper circuit typically converts the sine wave carrier into a square wave that is amplified and thus becomes the clock signal.

Balanced Modulators

A *balanced modulator* is a circuit that generates a DSB signal, suppressing the carrier and leaving only the two sidebands.

Lattice Modulator:

One of the most popular and widely used balanced modulators is the diode ring or *lattice modulator*.

It consists of an input transformer T_1 , an output transformer T_2 , and four diodes connected in a bridge circuit. The carrier signal is applied to the center taps of the input and output transformers, and the modulating signal is applied to the input transformer T_1 . The output appears across the secondary of the output transformer as shown in fig 1.29

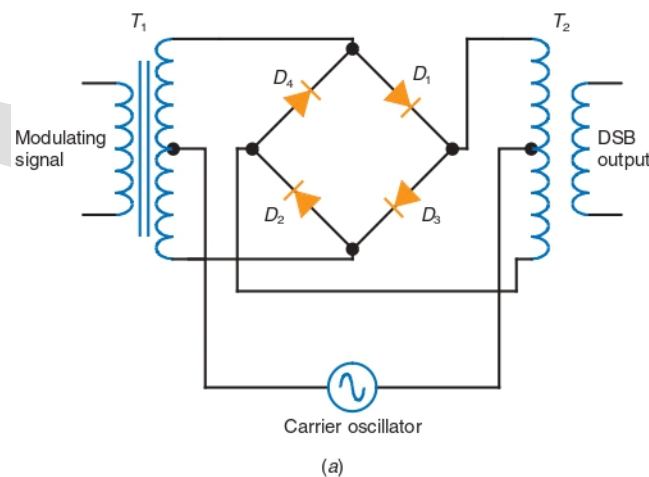


Fig 1,29 Lattice balanced Modulator

The operation of the circuit is explained by using the figure 1.30.

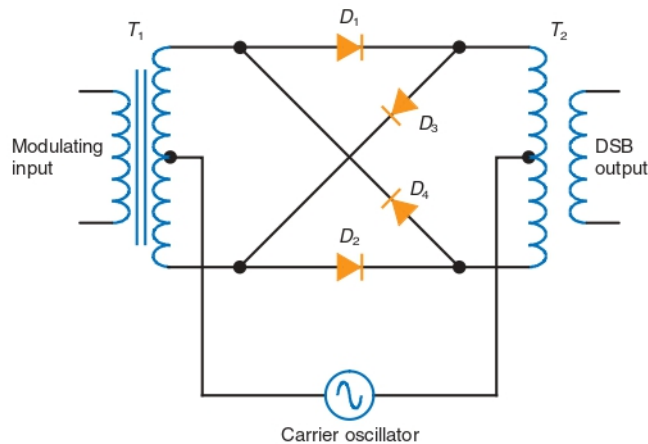


Figure 1.30 Lattice balanced modulator

Operation: The carrier sine wave, is used as a source of forward and reverse bias for the diodes. The carrier turns the diodes off and on at a high rate of speed, and the diodes act as switches that connect the modulating signal at the secondary of T_1 to the primary of T_2 .

If the modulating input is zero and the polarity of the carrier is positive, diodes D_1 and D_2 are forward-biased. At this time, D_3 and D_4 are reverse-biased and act as open circuits. Current divides equally in the upper and lower portions of the primary winding of T_2 . The current in the upper part of the winding produces a magnetic field that is equal and opposite to the magnetic field produced by the current in the lower half of the secondary. The magnetic fields thus cancel each other out. No output is induced in the secondary, and the carrier is effectively suppressed.

When the polarity of the carrier reverses, diodes D_1 and D_2 are reverse-biased and diodes D_3 and D_4 conduct. Again, the current flows in the secondary winding of T_1 and the primary winding of T_2 . The equal and opposite magnetic fields produced in T_2 cancel each other out. The carrier is effectively balanced out, and its output is zero.

When the message signal is applied to the primary of T_1 it appears across the secondary of T_1 . The diode switches connect the secondary of T_1 to the primary of T_2 at different times depending upon the carrier polarity. When the carrier polarity is positive diodes D_1 and D_2 conduct and act as closed switches. At this time, D_3 and D_4 are reverse-biased. As a result, the modulating signal at the secondary of T_1 is applied to the primary of T_2 through D_1 and D_2 .

When the carrier polarity reverses, D_1 and D_2 cut off and D_3 and D_4 conduct. Again, a portion of the modulating signal at the secondary of T_1 is applied to the primary of T_2 , but this time the leads have been effectively reversed because of the connections of D_3 and D_4 . The result is a 180° phase reversal. With this connection, if the modulating signal is positive, the output will be negative, and vice versa.

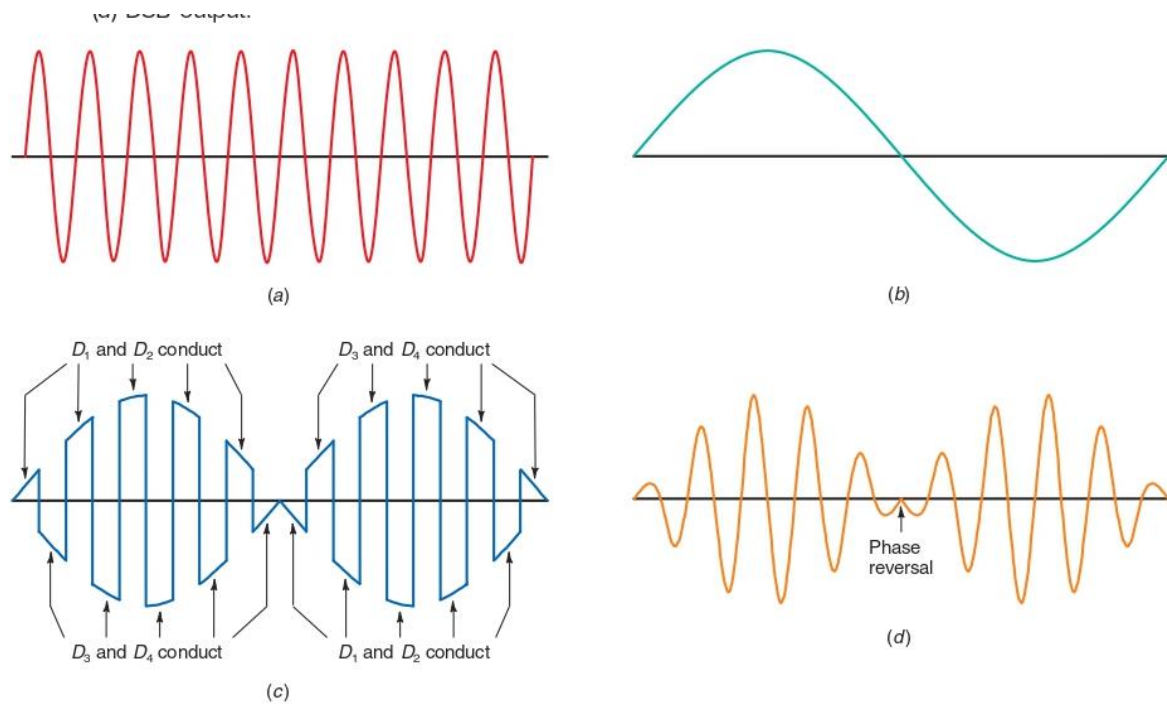


Figure 1.31 Waveforms in the lattice-type balanced modulator. (a) Carrier. (b) Modulating signal. (c) DSB signal—primary T_2 . (d) DSB output.

Frequency Division Multiplexing

In *frequency-division multiplexing (FDM)*, multiple signals share the bandwidth of a common communication channel. The FDM process divides up the bandwidth of the single channel into smaller, equally spaced channels, each capable of carrying information in sidebands

FDM Transmitter

Fig. 1.32 shows a general block diagram of an FDM transmitting system. Each signal to be transmitted feeds a modulator circuit. The carrier for each modulator (f_c) is on a different frequency. The carrier frequencies are usually equally spaced from one another over a specific frequency range. These carriers are referred to as *subcarriers*. The modulator outputs containing the sideband information are added algebraically in a linear mixer; no modulation or generation of sidebands takes place. The resulting output signal is a composite of all the modulated subcarriers. This signal can be used to modulate a radio transmitter or can itself be transmitted over the single communication channel.

Each input signal is given a portion of the bandwidth. The resulting spectrum is illustrated in Fig. 1.33

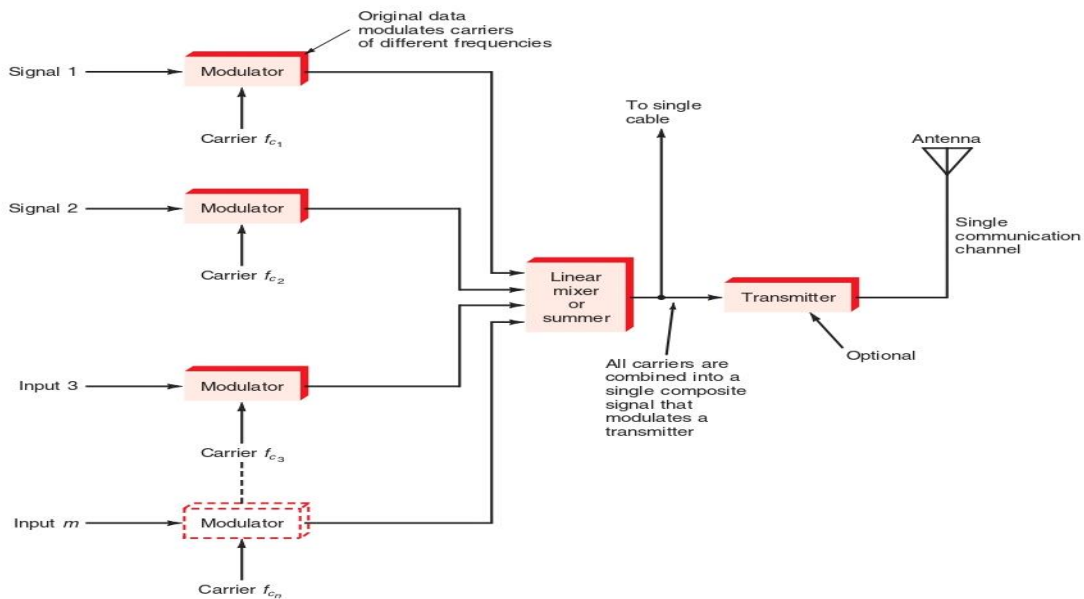


Figure 1.32 Transmitting end of FDM

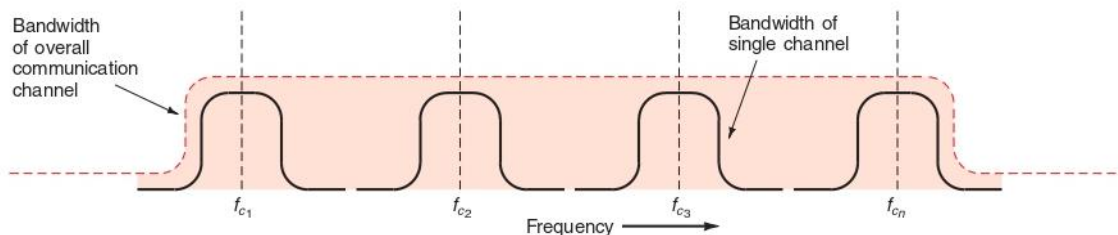


Figure 1.33 Spectrum of FDM

FDM Receiver:

The receiving portion of an FDM system is shown in Fig. 1-34. A receiver picks up the signal and demodulates it, recovering the composite signal. This is sent to a group of band pass filters, each centered on one of the carrier frequencies. Each filter passes only its channel and rejects all others. A channel demodulator then recovers each original input signal.

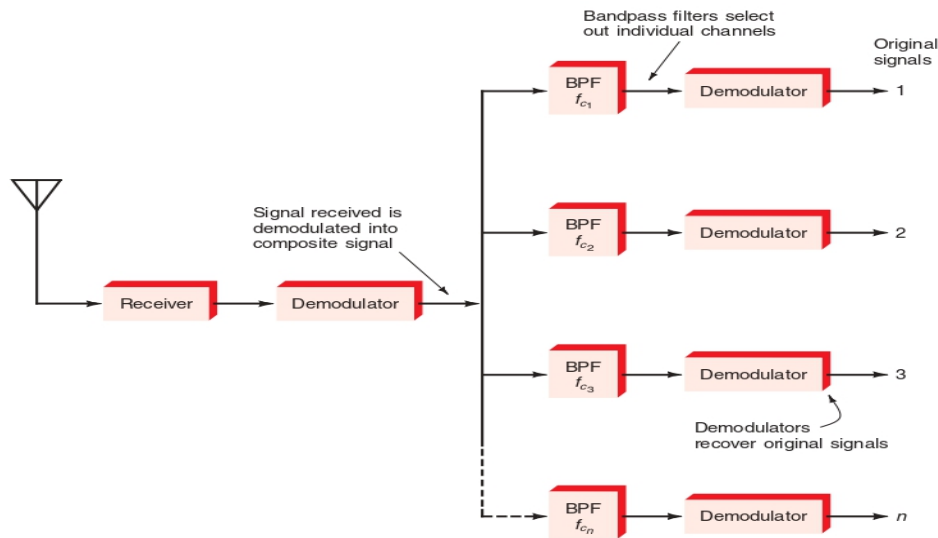


Figure 1.34 Receiving end of FDM

FDM applications

1. Telemetry

Sensors in telemetry systems generate electric signals that change in some way in response to changes in physical characteristics.

An example of a sensor is a *thermistor*, a device used to measure temperature. A thermistor's resistance varies inversely with temperature: As the temperature increases, the resistance decreases. The thermistor is usually connected into some kind of a resistive network, such as a voltage divider or bridge, and to a dc voltage source. The result is a dc output voltage, which varies in accordance with temperature and which is transmitted to a remote receiver for measurement, readout, and recording. The thermistor becomes one channel of an FDM system.

Other sensors have different kinds of outputs. Many have varying dc outputs, and others have ac output. Each of these signals is typically amplified, filtered, and otherwise conditioned before being used to modulate a carrier. All the carriers are then added to form a single multiplexed channel

Telemetry Transmitting end

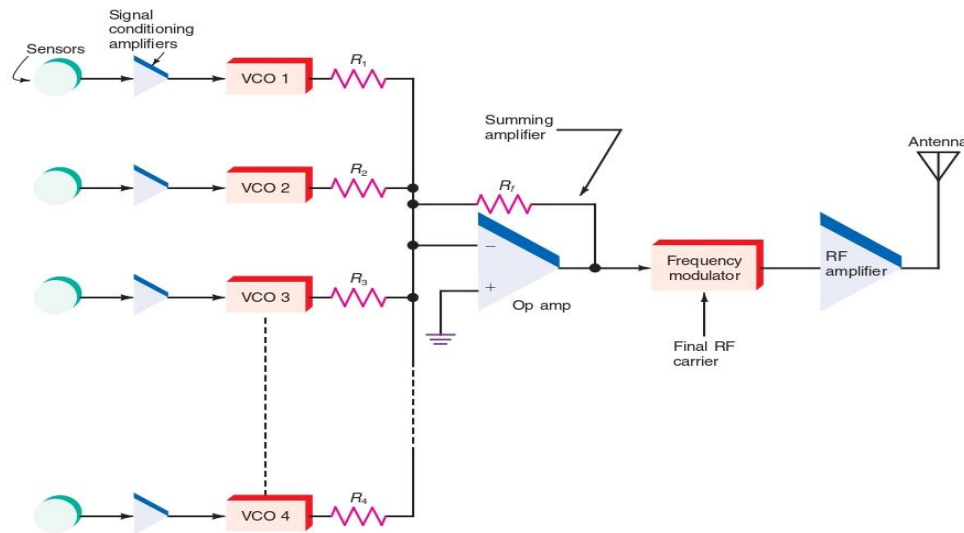


Figure 1.35 Telemetry transmitting system

The conditioned transducer outputs are normally used to frequency-modulate a subcarrier. The varying direct or alternating current output from sensor changes the frequency of the carrier. The oscillator used here is *voltage controlled oscillator (VCO)*. To produce FDM, each VCO operates at a different center or carrier frequency. The outputs of the subcarrier oscillators are added. A diagram of such a system is shown in Fig. 1-35

The linear mixing process in an FDM system can be accomplished with a simple resistor network. However, such networks greatly attenuate the signal, and some voltage amplification is usually required for practical systems. A way to achieve the mixing and amplification at the same time is to use an op amp summer

The composite output signal is then typically used to modulate a radio transmitter. A system that uses FM of the VCO subcarriers as well as FM of the final carrier is usually called an *FM/FM system*.

Telemetry receiving end:

The receiving end of a telemetry system is shown in Fig. 1-36. A standard superheterodyne receiver tuned to the RF carrier frequency is used to pick up the signal. An FM demodulator reproduces the original composite multiplexed signal, which is then fed to a demultiplexer. The demultiplexer divides the signals and reproduces the original inputs.

The output of the first FM demodulator is fed simultaneously to multiple bandpass filters, each of which is tuned to the center frequency of one of the specified subchannels. Each filter passes only its subcarrier and related sidebands and rejects all the others. The demultiplexing process is, then, essentially one of using filters to sort the composite multiplex signal back into its original components. The output of each filter is the subcarrier oscillator frequency with its modulation.

These signals are then applied to FM demodulators. Also known as *discriminators*, these circuits take the FM signal and recreate the original dc or ac signal produced by the transducer. The original signals are then measured or processed to provide the desired information from the remote transmitting source.

FDM telemetry systems, which are inexpensive and highly reliable, are still widely used in aircraft and missile instrumentation and for monitoring of medical devices, such as pacemakers.

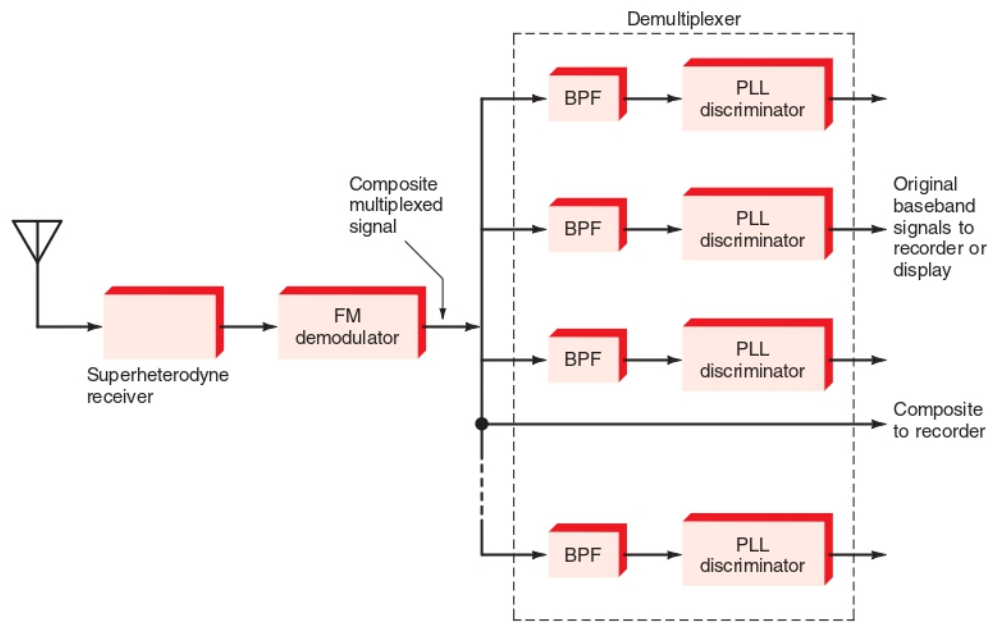


Figure 1.36 Telemetry receiving system

Cable TV. One of the best examples of FDM is cable TV, in which multiple TV signals, each in its own 6-MHz channel, are multiplexed on a common coaxial or fiber-optic cable and sent to nearby homes. TV signals include video and audio to modulate carriers using analog methods. Each channel uses a separate set of carrier frequencies, which can be added to produce FDM. The cable box acts as a tunable filter to select the desired channel.

Fig. 1-37 shows the spectrum on the cable. Each 6-MHz channel carries the video and voice of the TV signal. Coaxial and fiber-optic cables have an enormous bandwidth and can carry more than one hundred TV channels. Many cable TV companies also use their cable system for Internet access. A special modem (modulator-demodulator) permits computer data to be transmitted and received at very high data speeds.

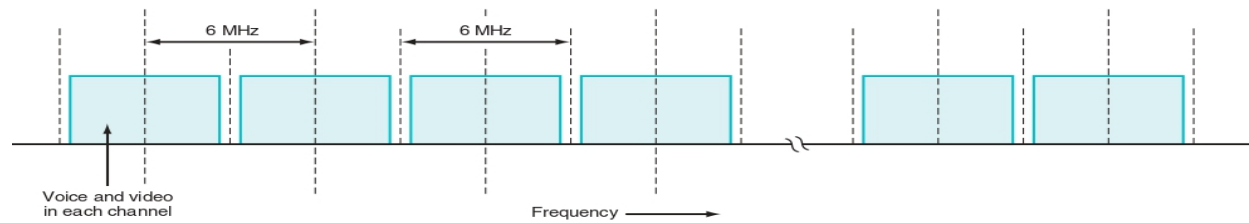


Figure 1.37 The spectrum on coaxial cable in a cable TV system with 6-MHz-wide channels.

FM Stereo Broadcasting. In recording original stereo, two microphones are used to generate two separate audio signals. The two microphones pick up sound from a common source, such as a voice or an orchestra, but from different directions. The separation of the two microphones provides sufficient differences in the two audio signals to provide more realistic reproduction of the original sound.

When stereo is reproduced, the two signals can come from a cassette tape, a CD, or some other source. These two independent signals must somehow be transmitted by a single transmitter. This is done through FDM techniques.

Fig. 1-38 is a general block diagram of a stereo FM multiplex modulator. The two audio signals, generally called the left (L) and right (R) signals, originate at the two microphones shown in the figure. These two signals are fed to a combining circuit, where they are used to form sum ($L + R$) and difference ($L - R$) signals. The $L + R$ signal is a linear algebraic combination of the left and right channels. The composite signal it produces is the same as if a single microphone were used to pick up the sound. It is this signal that a monaural receiver will hear.

The combining circuit inverts the right channel signal, thereby subtracting it from the left channel to produce the $L - R$ signal. These two signals, the $L + R$ and $L - R$, are transmitted independently and recombined later in the receiver to produce the individual right- and left-hand channels.

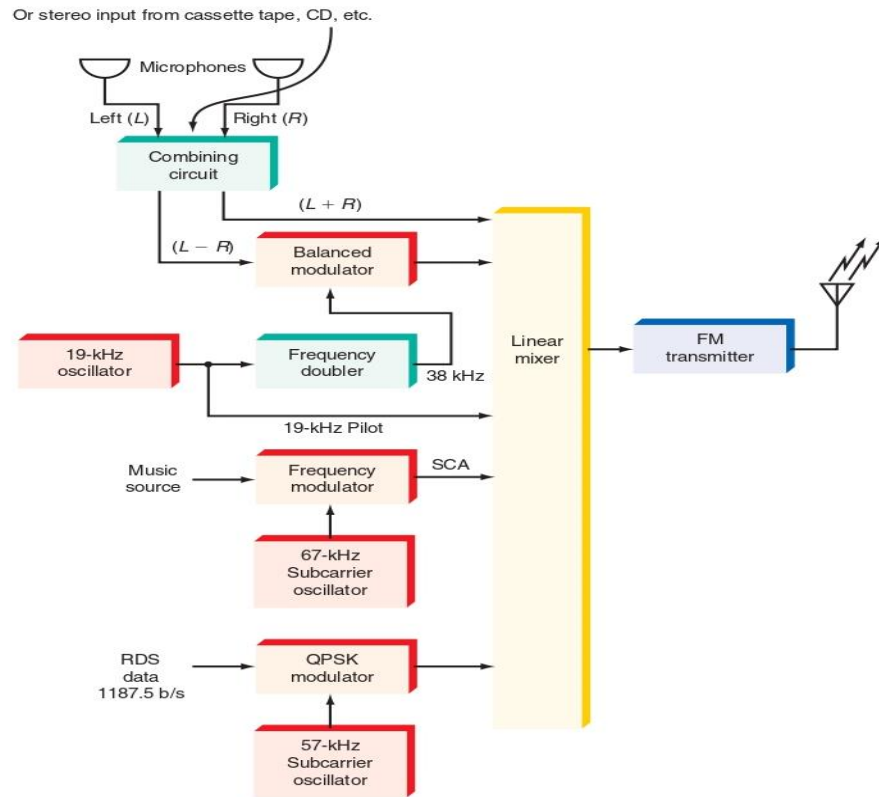


Figure 1.38 FM Stereo multiplexer

At the receiving end, the demodulation is accomplished with a circuit similar to that illustrated in Fig. 1-39. The FM superheterodyne receiver picks up the signal, amplifies it, and translates it to an intermediate frequency, usually 10.7 MHz. It is then demodulated. The output of the demodulator is the original multiplexed signal. The additional circuits now sort out the various signals and reproduce them in their original form. The original audio $L + R$ signal is extracted simply by passing the multiplex signal through a low-pass filter.

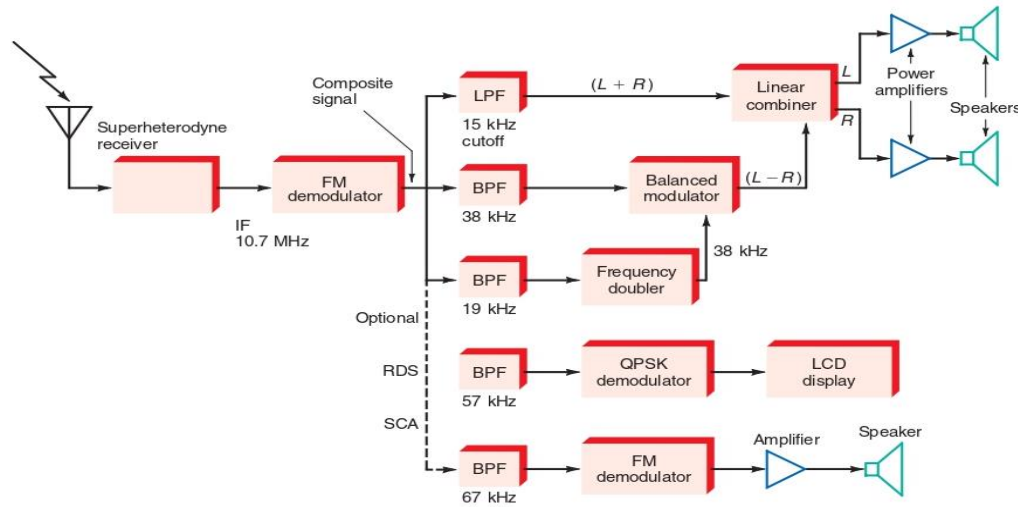


Figure 1.39 Recovering FM Stereo

The multiplexed signal is also applied to a bandpass filter that passes the 38-kHz suppressed subcarrier with its sidebands. This is the $L + R$ signal that modulates the 38-kHz carrier. This signal is fed to a balanced modulator for demodulation.

The 19-kHz pilot carrier is extracted by passing the multiplexed signal through a narrow bandpass filter. This 19-kHz subcarrier is then fed to an amplifier and frequency doubler circuit, which produces a 38-kHz carrier signal that is fed to the balanced modulator. The output of the balanced modulator, is the $L - R$ audio signal. This is fed to the linear resistive combiner along with the $L + R$ signal

The linear combiner both adds and subtracts these two signals. Addition produces the left-hand channel: $(L + R) + (L - R) = 2L$. Subtraction produces the right-hand channel: $(L + R) - (L - R) = 2R$. The left- and right-hand audio signals are then sent to separate audio amplifiers and ultimately the speaker